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Singh et al.

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(54) **OUTDOOR AIR HOOD ASSEMBLY WITH AN INLET HOOD**

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F24F 1/48 (2011.01)

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CPC . **F24F 1/58** (2013.01); **F24F 1/48** (2013.01)

(58) **Field of Classification Search**
CPC F24F 1/48; F24F 1/58; F24F 13/28
See application file for complete search history.

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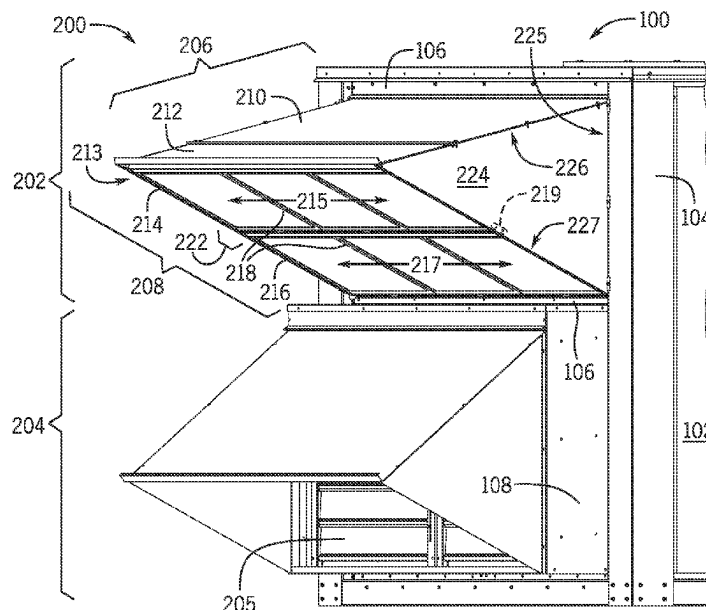
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(57) **ABSTRACT**

A hood assembly for a heating, ventilation, and air conditioning (HVAC) unit includes a top panel comprising a first panel portion and a second panel portion adjustably coupled to one another, wherein the top panel is configured to rotatably couple to a housing of the HVAC unit. The hood assembly further includes a filter frame rotatably coupled to the top panel, wherein the filter frame is configured to support at least one filter. The hood assembly is adjustable between a collapsed configuration and a deployed configuration, the first panel portion and the second panel portion are configured to translate relative to one another during transition of the hood assembly between the collapsed configuration and the deployed configuration, and the top panel is configured to contain the filter frame within the housing in the collapsed configuration.

19 Claims, 12 Drawing Sheets



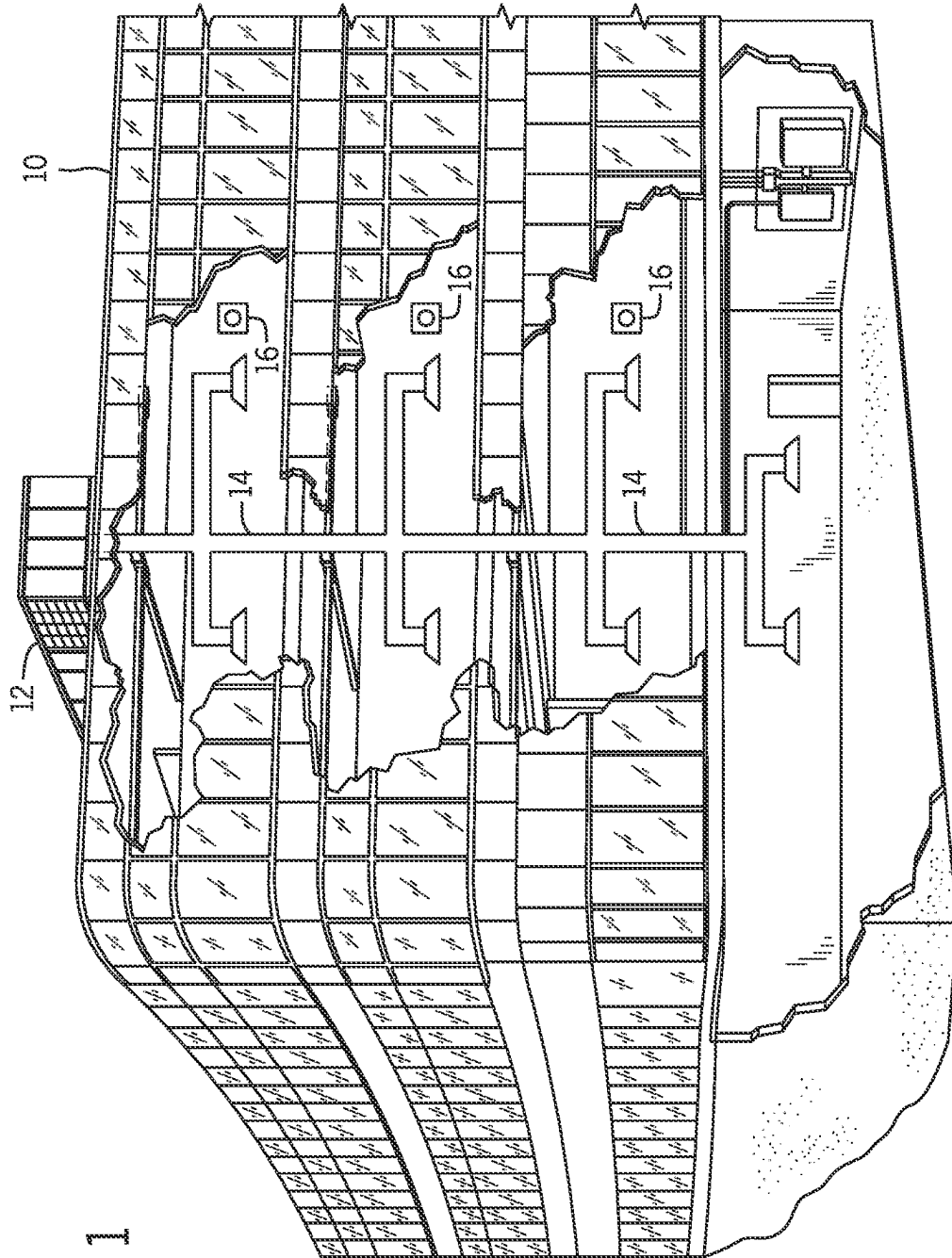
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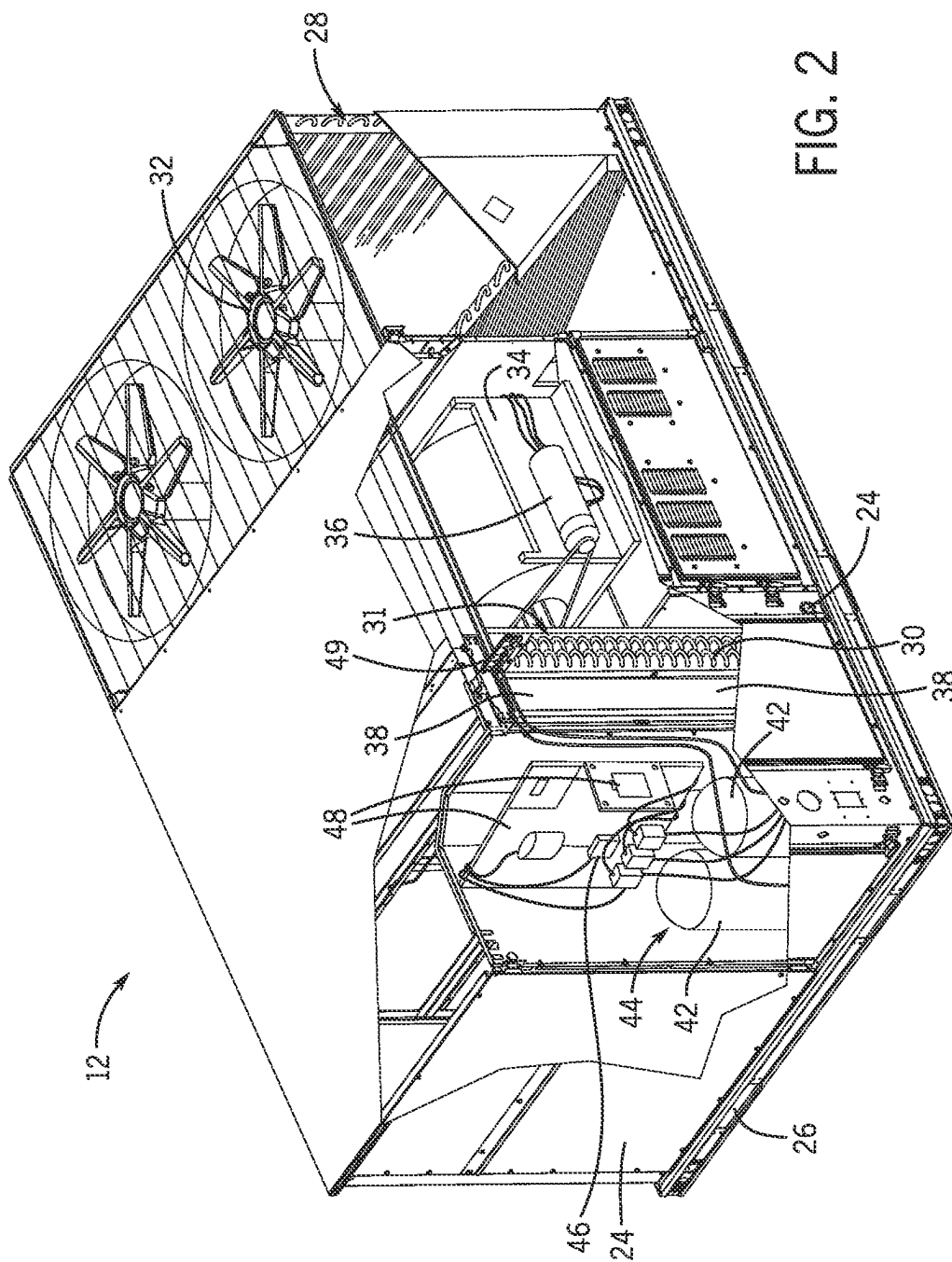


FIG. 2

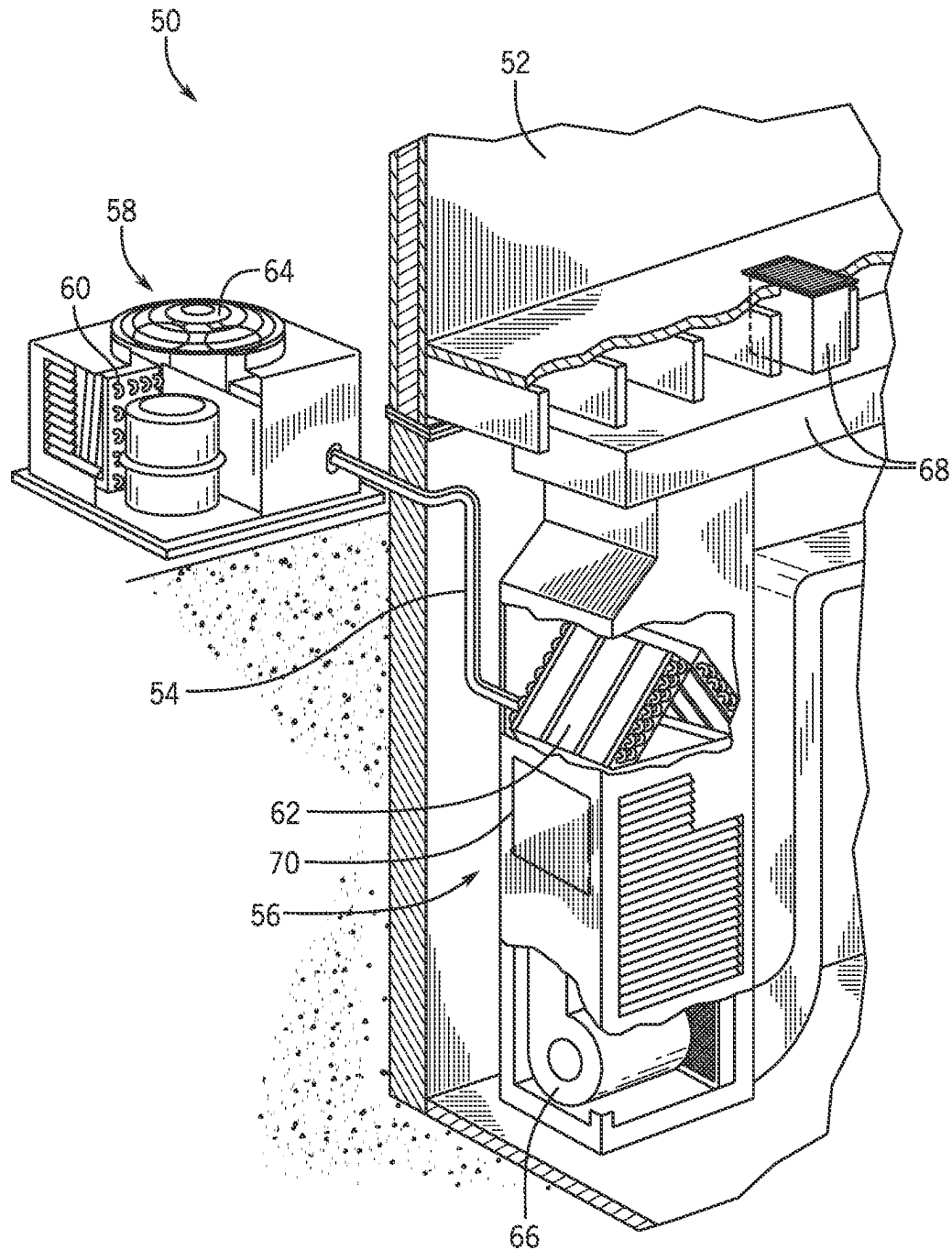


FIG. 3

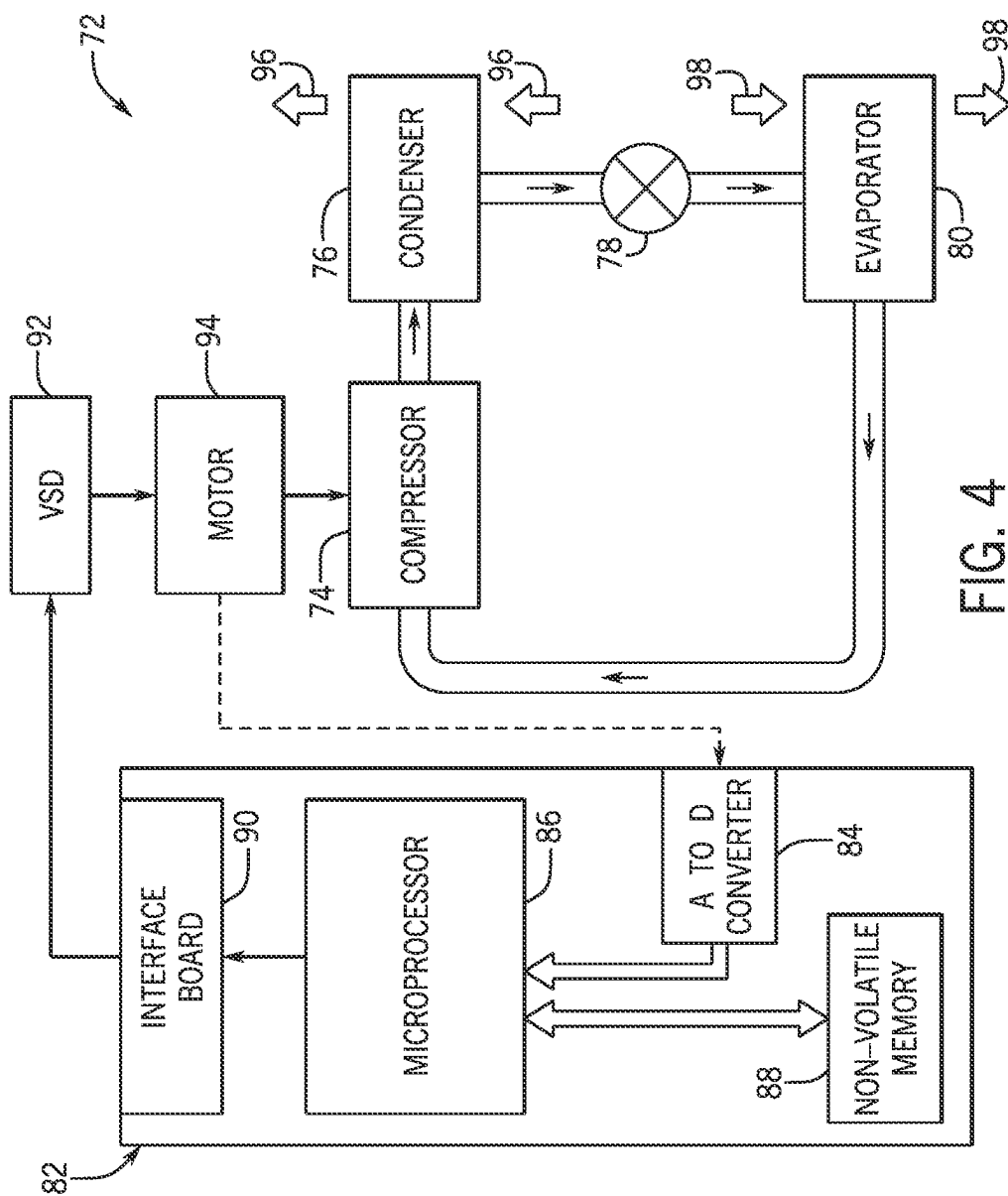


FIG. 4

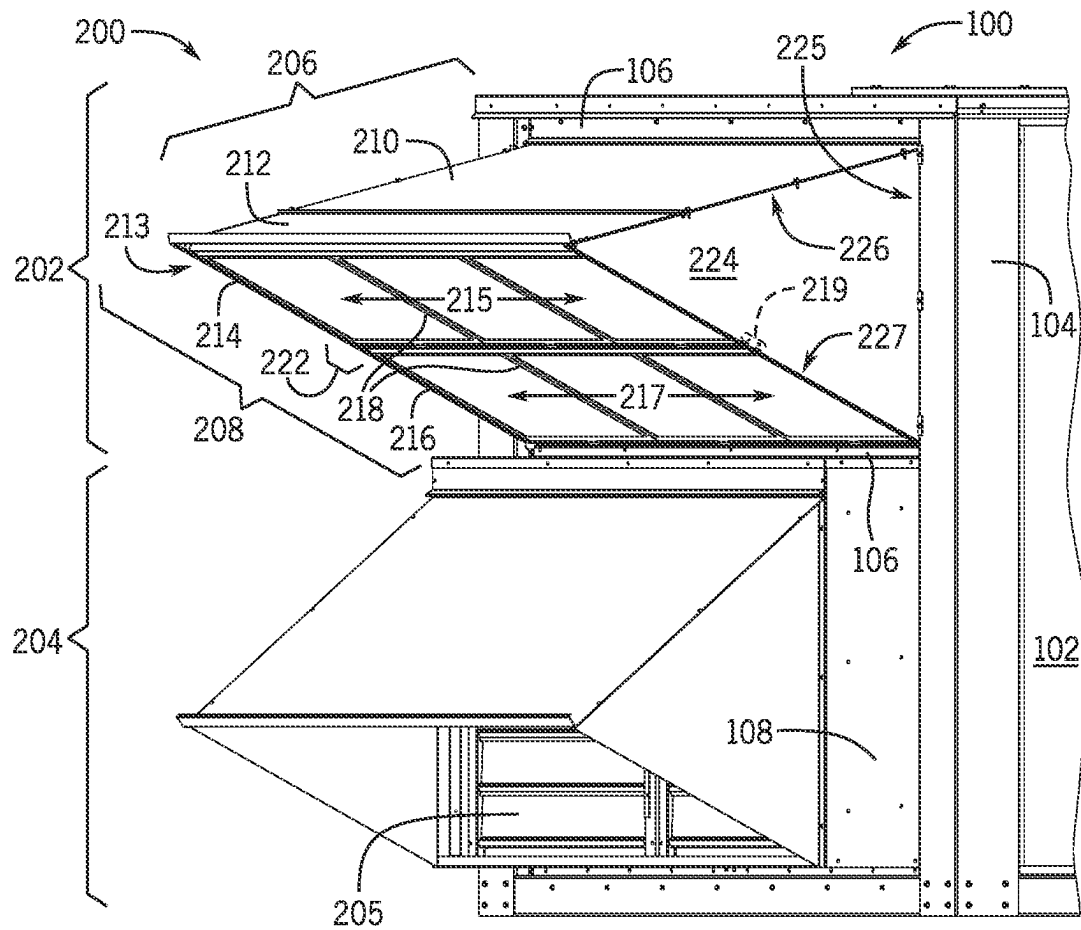


FIG. 5

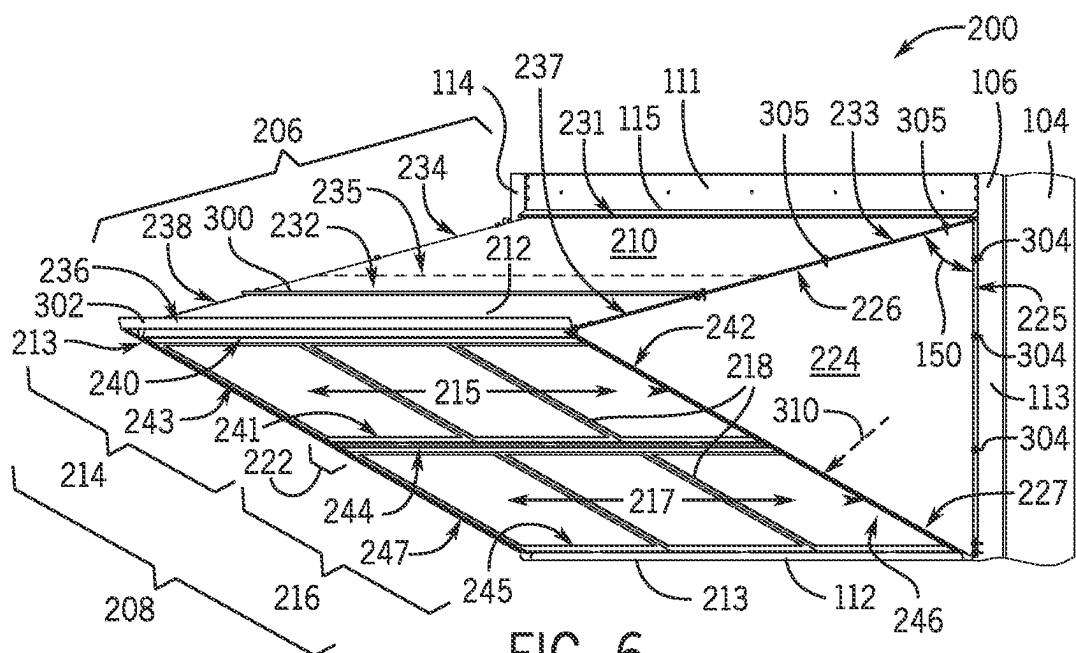


FIG. 6

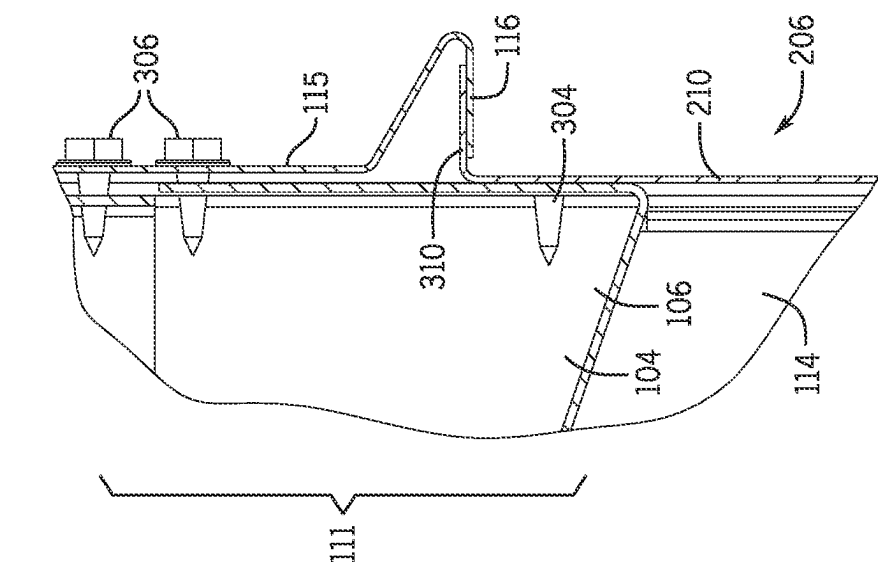


FIG. 7B

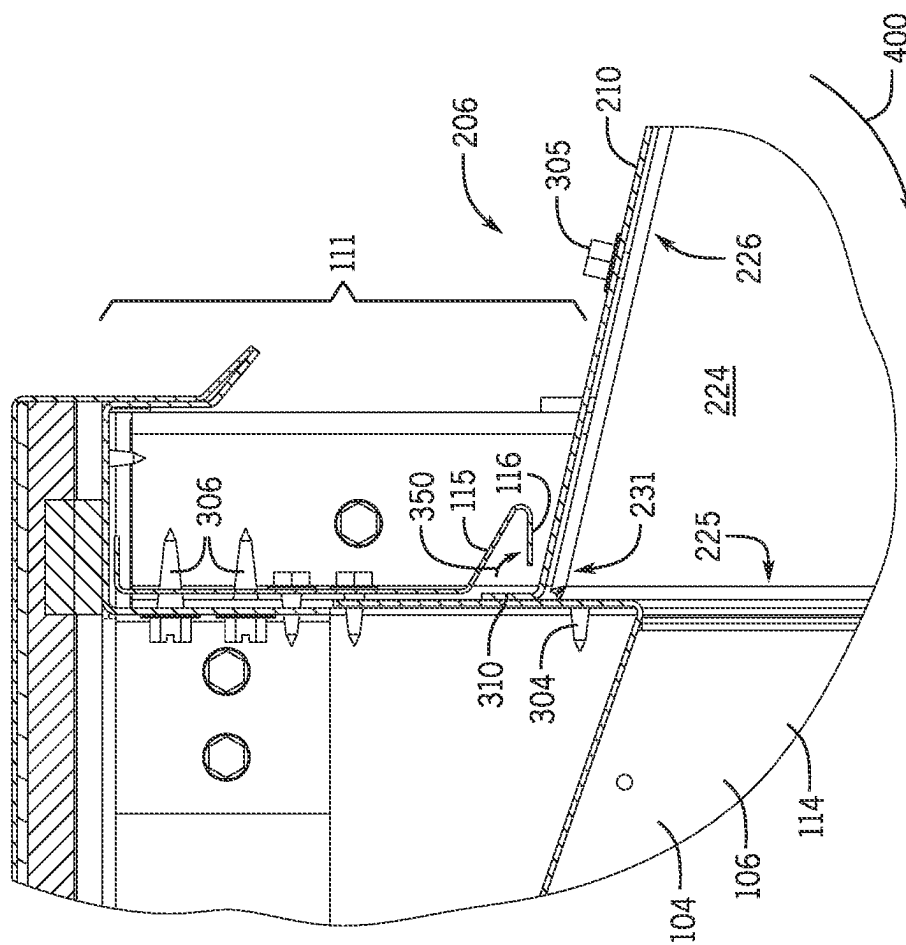


FIG. 7A

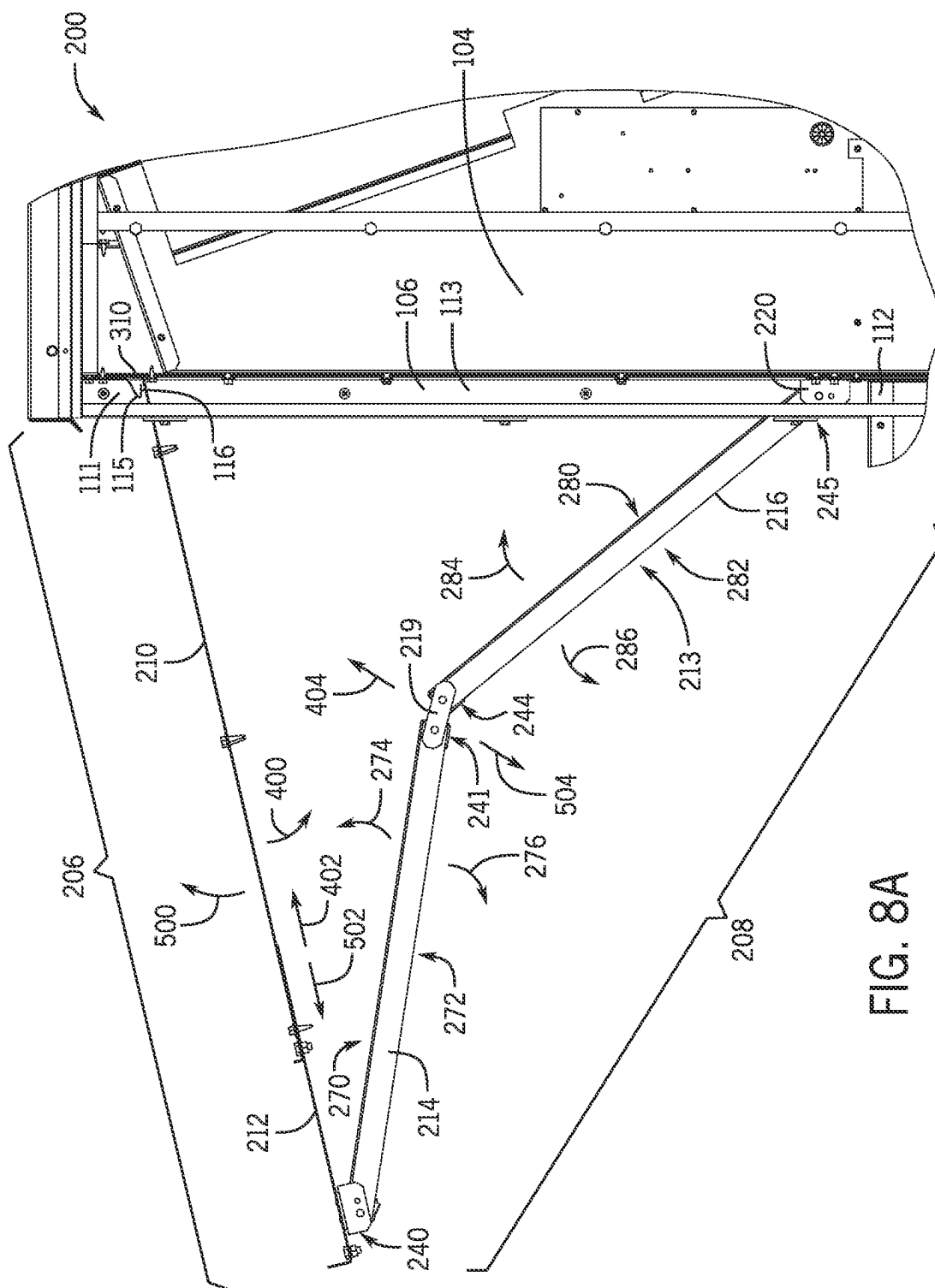


FIG. 8A

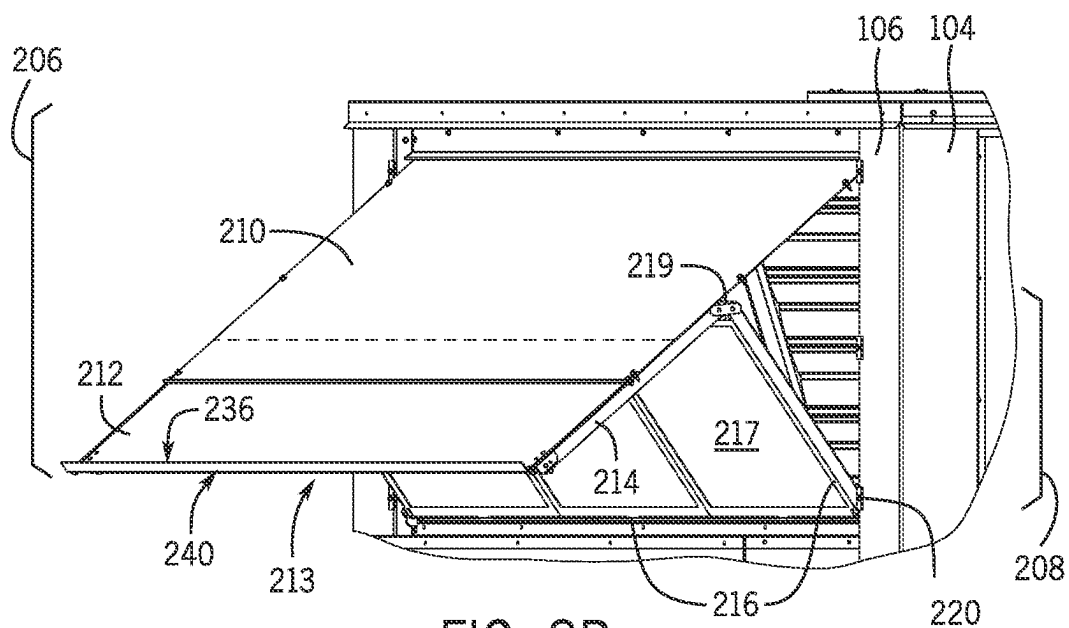


FIG. 8B

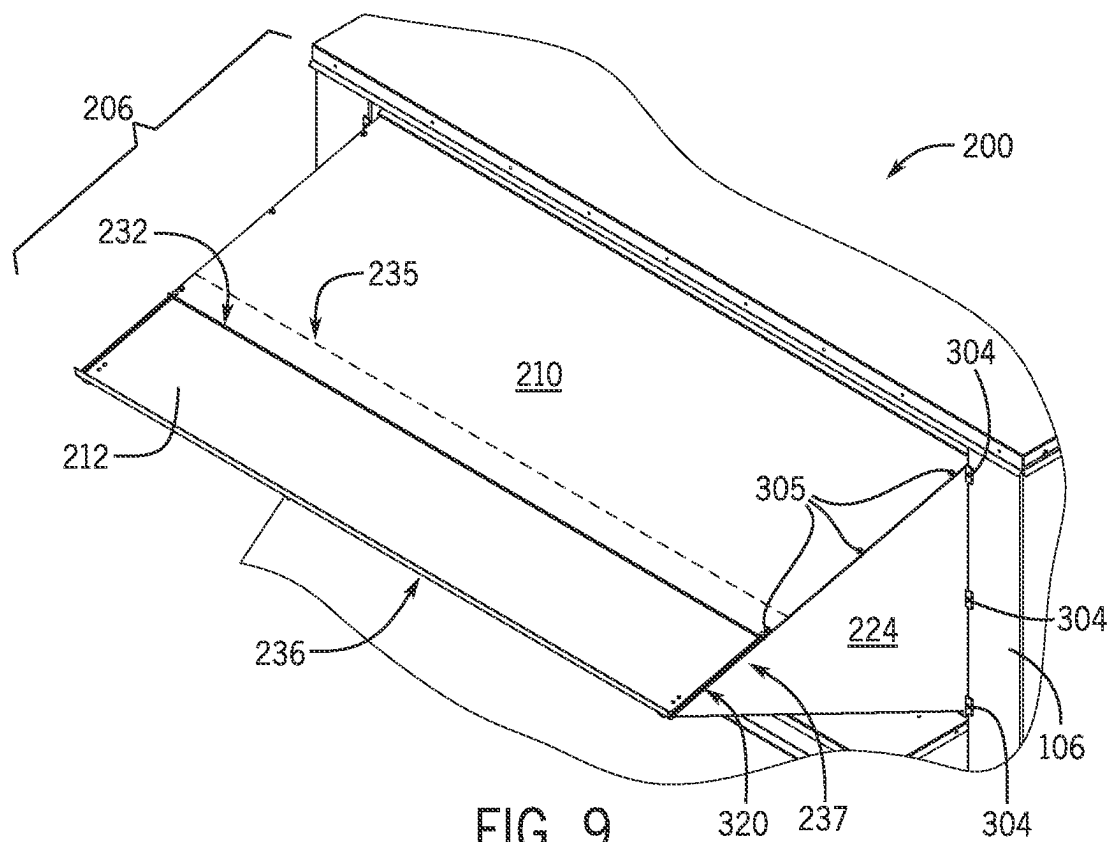


FIG. 9

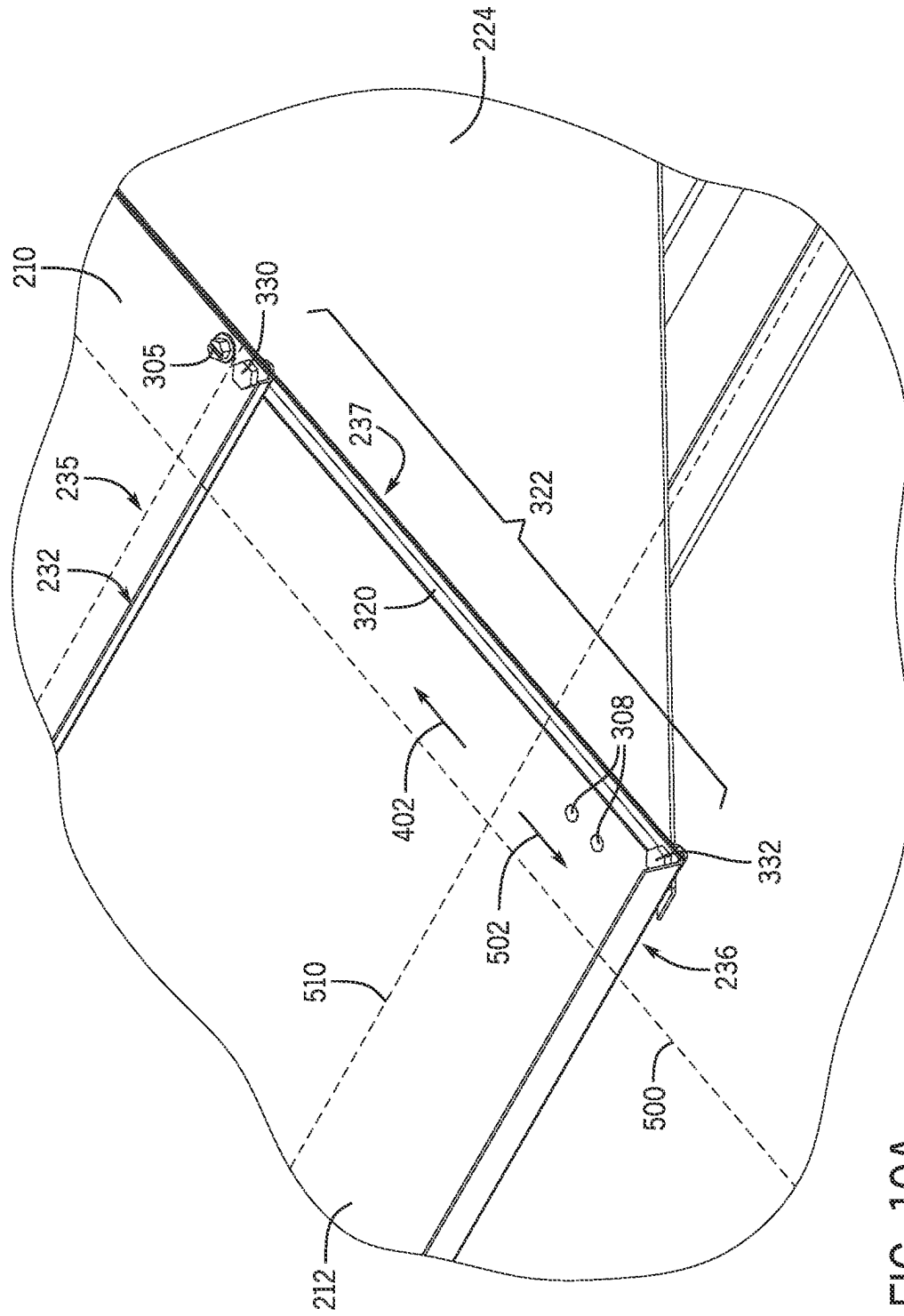


FIG. 10A

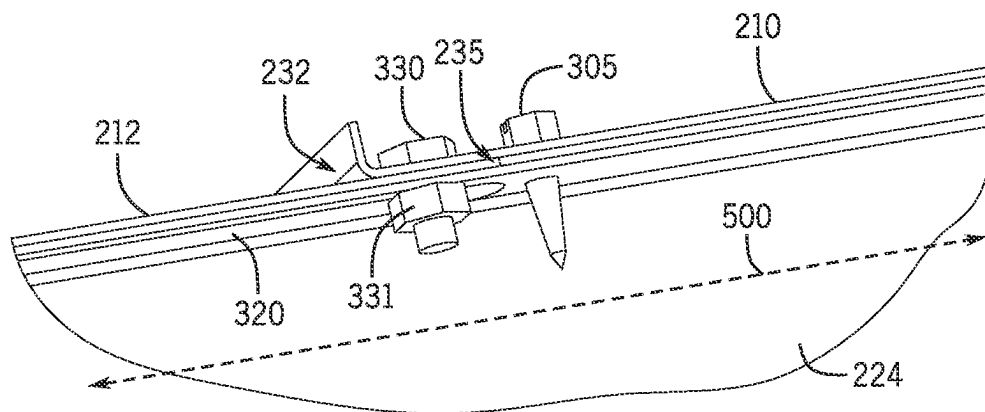


FIG. 10B

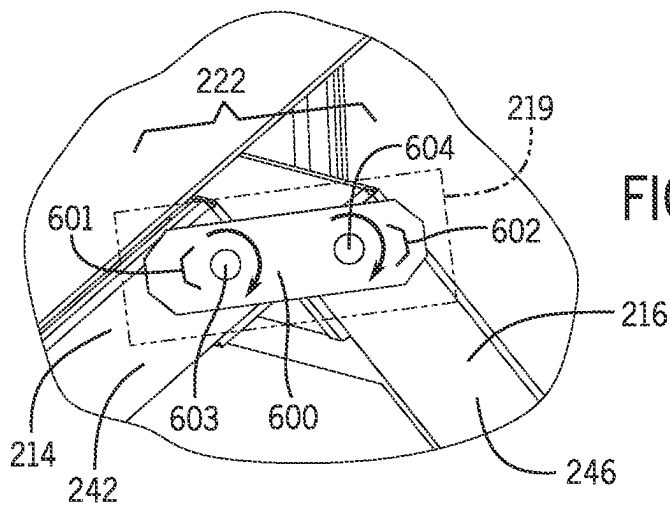


FIG. 11A

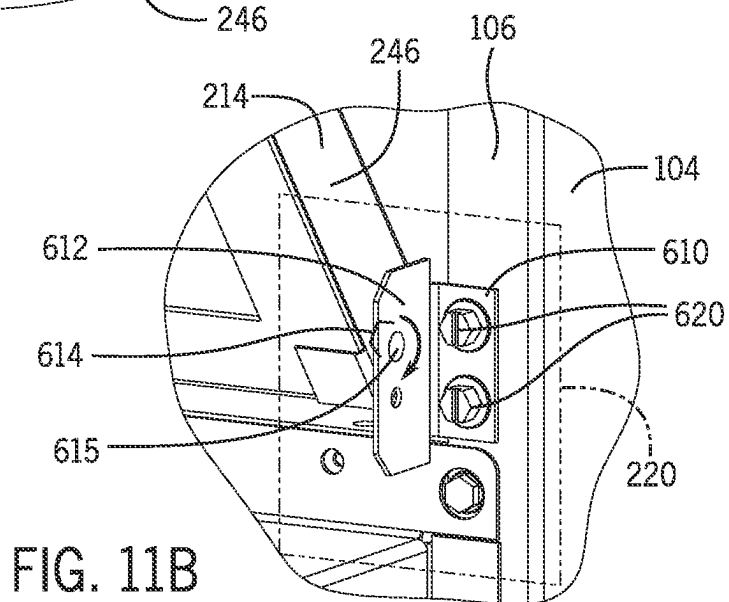


FIG. 11B

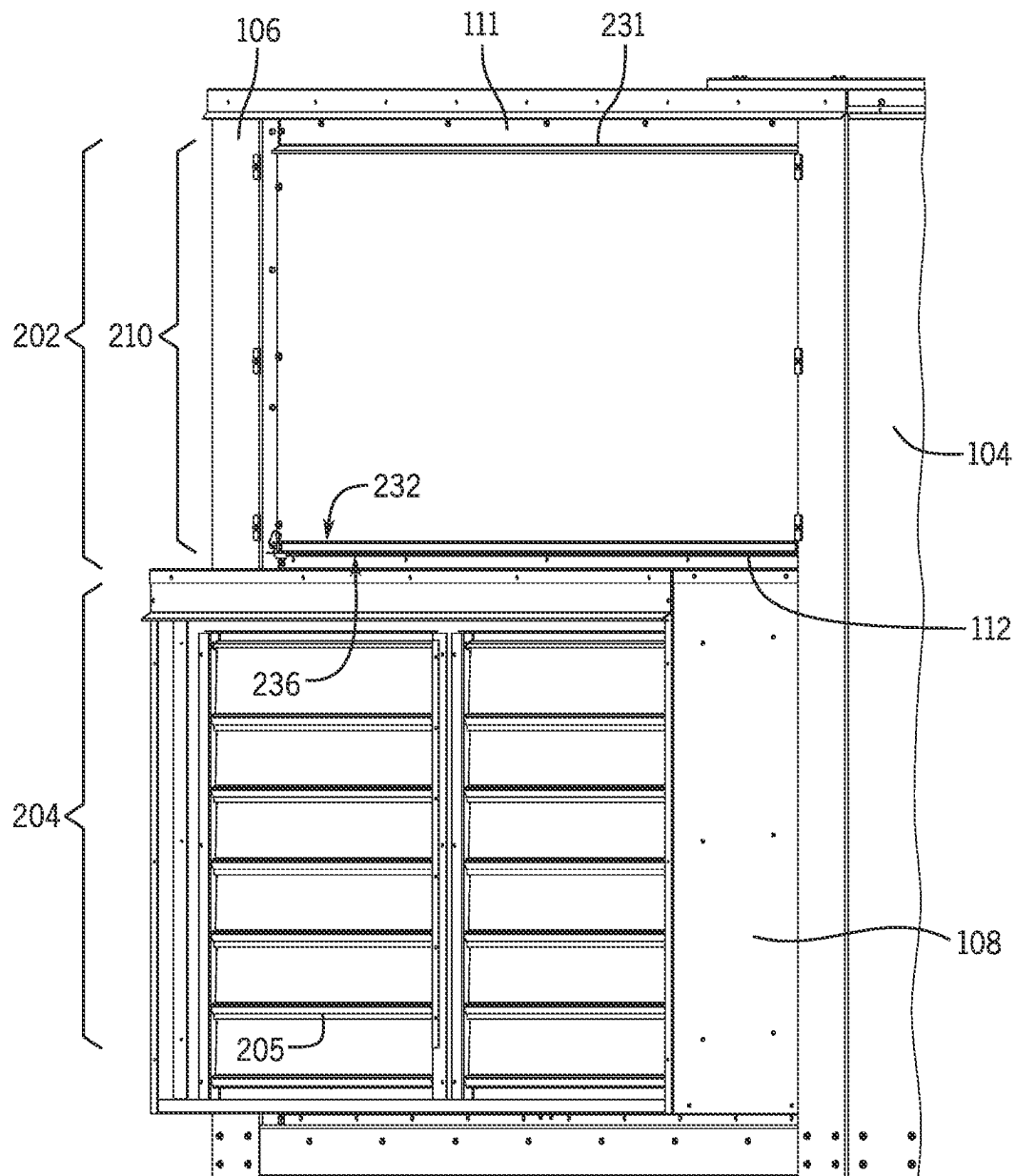


FIG. 12

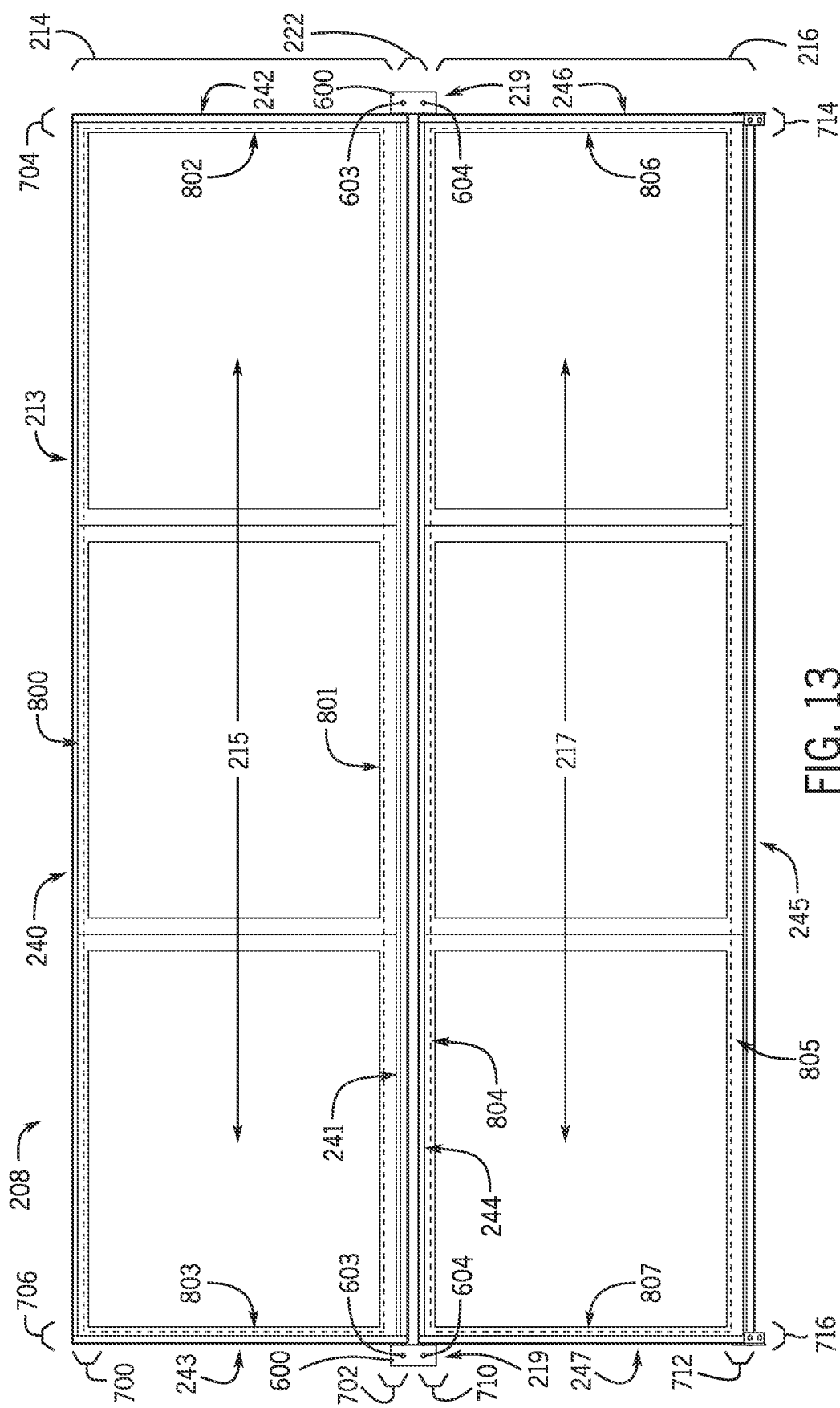


FIG. 13

OUTDOOR AIR HOOD ASSEMBLY WITH AN INLET HOOD**CROSS REFERENCE TO RELATED APPLICATION**

This application is a continuation of U.S. patent application Ser. No. 17/235,726, entitled "AN OUTDOOR AIR HOOD ASSEMBLY WITH AN INLET HOOD," filed Apr. 20, 2021, which claims priority from and the benefit of India Provisional Application Serial No. 202011016873, entitled "AN OUTDOOR AIR HOOD ASSEMBLY WITH AN INLET HOOD," filed Apr. 20, 2020, each of which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure and are described below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be noted that these statements are to be read in this light, and not as admissions of prior art.

Heating, ventilation, and air conditioning (HVAC) systems are utilized to control environmental properties, such as temperature and humidity, for occupants of residential, commercial, and industrial environments. The HVAC systems may control the environmental properties through control of an air flow delivered to the environment. In some cases, the HVAC systems include an intake hood that is configured to block water, dust, debris, and other contaminants from entering the HVAC system. For instance, an outdoor HVAC unit may draw air into the unit from the intake hood, and the air may ultimately flow across a heat exchanger to exchange thermal energy with a working fluid (e.g., refrigerant) flowing through the heat exchanger. Existing intake hoods are generally shipped separately from a housing of the HVAC unit and are therefore assembled upon delivery, which increases transportation, delivery, and installation costs and may be time-consuming. Accordingly, it is now recognized that improved assembly and transportation management for HVAC units having intake hoods is desired.

SUMMARY

A summary of certain embodiments disclosed herein is set forth below. It should be noted that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

In an embodiment, a hood assembly for a heating, ventilation, and air conditioning (HVAC) unit comprises a top panel comprising a first panel portion and a second panel portion adjustably coupled to one another, wherein the top panel is configured to rotatably couple to a housing of the HVAC unit. The HVAC unit further comprises a filter frame rotatably coupled to the top panel, wherein the filter frame is configured to support at least one filter. The hood assembly is adjustable between a collapsed configuration and a deployed configuration, the first panel portion and the second panel portion are configured to translate relative to one another during transition of the hood assembly between the collapsed configuration and the deployed configuration, and

the top panel is configured to contain the filter frame within the housing in the collapsed configuration.

In another embodiment, a heating, ventilation, and air conditioning (HVAC) unit comprises a housing and an intake hood assembly coupled to the housing and configured to transition between a collapsed configuration and a deployed configuration. The HVAC unit further comprises a top cover of the intake hood rotatably coupled to the housing and comprising a first panel and a second panel, wherein the second panel is configured to translate relative to the first panel during transition of the intake hood assembly between the collapsed configuration and the deployed configuration. The HVAC unit further comprises a filter frame rotatably coupled to the top cover and configured to support at least one filter, wherein the top cover and the filter frame are configured to rotate relative to one another and relative to the housing during transition of the intake hood assembly between the collapsed configuration and the deployed configuration.

In another embodiment, an intake hood assembly for a heating, ventilation, and air conditioning (HVAC) unit includes a top panel comprising a first panel portion and a second panel portion adjustably coupled to one another, wherein the top panel is configured to couple to a housing of the HVAC unit. The intake hood assembly further includes a filter frame coupled to the top panel, wherein the filter frame comprises a first sub-frame configured to support a first filter and a second sub-frame configured to support a second filter. The intake hood assembly is adjustable between a collapsed configuration and a deployed configuration, the first panel portion and the second panel portion are configured to translate relative to one another during transition of the intake hood assembly between the collapsed configuration and the deployed configuration, and the first sub-frame and the second sub-frame are configured to rotate relative to one another during transition of the intake hood assembly between the collapsed configuration and the deployed configuration.

BRIEF DESCRIPTION OF THE DRAWINGS

Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:

FIG. 1 is a perspective view of a building having an embodiment of a heating, ventilation, and air conditioning (HVAC) system for environmental management that may employ one or more HVAC units, in accordance with an aspect of the present disclosure;

FIG. 2 is a perspective view of an embodiment of a packaged HVAC unit that may be used in the HVAC system of FIG. 1, in accordance with an aspect of the present disclosure;

FIG. 3 is a cutaway perspective view of an embodiment of a residential, split HVAC system, in accordance with an aspect of the present disclosure;

FIG. 4 is a schematic illustration of an embodiment of a vapor compression system that can be used in any of the systems of FIGS. 1-3, in accordance with an aspect of the present disclosure;

FIG. 5 is a perspective view of an embodiment of a portion of an HVAC unit having an air hood assembly in a deployed configuration, in accordance with an aspect of the present disclosure;

FIG. 6 is a perspective view of an embodiment of an air hood assembly in a deployed configuration, in accordance with an aspect of the present disclosure;

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FIG. 7A is a cross-sectional side view of an embodiment of a top panel retainer of an air hood assembly in a deployed configuration, in accordance with an aspect of the present disclosure;

FIG. 7B is a cross-sectional side view of an embodiment of a top panel retainer of an air hood assembly in a collapsed configuration, in accordance with an aspect of the present disclosure;

FIG. 8A is a schematic side view illustrating transition of an embodiment of an air hood assembly between a deployed configuration and a collapsed configuration, in accordance with an aspect of the present disclosure;

FIG. 8B is a perspective view of an embodiment of an air hood assembly, illustrating transition between a deployed configuration and a collapsed configuration, in accordance with an aspect of the present disclosure;

FIG. 9 is a perspective view of an embodiment of an air hood assembly, in accordance with an aspect of the present disclosure;

FIG. 10A is an expanded perspective view of an embodiment of an air hood assembly, illustrating panel portions of the air hood assembly, in accordance with an aspect of the present disclosure;

FIG. 10B is an expanded perspective view of an embodiment of an air hood assembly, illustrating coupled panel portions of the air hood assembly, in accordance with an aspect of the present disclosure;

FIG. 11A is a perspective view of an embodiment of a connection between filter frames of an air hood assembly, in accordance with an aspect of the present disclosure;

FIG. 11B is a perspective view of an embodiment of a connection between a filter frame of an air hood assembly and a housing frame, in accordance with an aspect of the present disclosure;

FIG. 12 is a perspective view of an embodiment of an air hood assembly in a collapsed configuration, in accordance with an aspect of the present disclosure; and

FIG. 13 is a schematic of an embodiment of a filter assembly of an air hood assembly, in accordance with an aspect of the present disclosure.

DETAILED DESCRIPTION

One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be noted that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be noted that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

When introducing elements of various embodiments of the present disclosure, the articles "a," "an," and "the" are intended to mean that there are one or more of the elements. The terms "comprising," "including," and "having" are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be noted that references to "one embodiment" or "an embodiment" of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

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The present disclosure is directed to an improved hood assembly (e.g., air hood assembly, air intake hood assembly) configured to be disposed over an air intake (e.g., air inlet) of a heating, ventilation, and air conditioning (HVAC) unit. As discussed above, existing air hoods may be removed and/or otherwise disassembled to an HVAC unit when the HVAC unit is to be transported from one location to another. As such, the HVAC unit and the air hood are shipped to a new location separately from one another, which may increase transportation costs. The air hood must also be assembled, or otherwise attached, to the HVAC unit when the separate components reach the final destination, which may be time-consuming and costly.

Accordingly, embodiments of the present disclosure are directed to a collapsible air hood assembly that is configured to transition between a collapsed configuration and a deployed configuration. In some embodiments, the air hood assembly may include a top panel having a first panel and a second panel and having a pair of side panels removably coupled to the top panel. When the air hood assembly is in the deployed configuration, the air hood assembly is configured to block water and other elements, such as dust, debris, and/or dirt particles present in an environment surrounding the HVAC unit, from entering the HVAC unit via an air intake, while also enabling air to flow from the environment surrounding the HVAC unit into the HVAC unit via the air intake. When the air hood assembly is in the collapsed configuration, the pair of side panels may be removed such that the top panel of the air hood assembly may transition and rest generally flush with a side of the HVAC unit. As such, the air hood assembly may be transported with the HVAC unit in an assembled or installed configuration without substantially increasing the size (e.g., footprint) of the HVAC unit, thereby reducing the time and costs associated with transportation, delivery, and assembly of the HVAC unit.

Turning now to the drawings, FIG. 1 illustrates a heating, ventilation, and air conditioning (HVAC) system for building environmental management that may employ one or more HVAC units. As used herein, an HVAC system includes any number of components configured to enable regulation of parameters related to climate characteristics, such as temperature, humidity, air flow, pressure, air quality, and so forth. For example, an "HVAC system" as used herein is defined as conventionally understood and as further described herein. Components or parts of an "HVAC system" may include, but are not limited to, all, some of, or individual parts such as a heat exchanger, a heater, an air flow control device, such as a fan, a sensor configured to detect a climate characteristic or operating parameter, a filter, a control device configured to regulate operation of an HVAC system component, a component configured to enable regulation of climate characteristics, or a combination thereof. An "HVAC system" is a system configured to provide such functions as heating, cooling, ventilation, dehumidification, pressurization, refrigeration, filtration, or any combination thereof. The embodiments described herein may be utilized in a variety of applications to control climate characteristics, such as residential, commercial, industrial, transportation, or other applications where climate control is desired.

In the illustrated embodiment, a building 10 is air conditioned by a system that includes an HVAC unit 12. The building 10 may be a commercial structure or a residential structure. As shown, the HVAC unit 12 is disposed on the roof of the building 10; however, the HVAC unit 12 may be located in other equipment rooms or areas adjacent the

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building 10. The HVAC unit 12 may be a single package unit containing other equipment, such as a blower, integrated air handler, and/or auxiliary heating unit. In other embodiments, the HVAC unit 12 may be part of a split HVAC system, such as the system shown in FIG. 3, which includes an outdoor HVAC unit 58 and an indoor HVAC unit 56.

The HVAC unit 12 is an air cooled device that implements a refrigeration cycle to provide conditioned air to the building 10. Specifically, the HVAC unit 12 may include one or more heat exchangers across which an air flow is passed to condition the air flow before the air flow is supplied to the building. In the illustrated embodiment, the HVAC unit 12 is a rooftop unit (RTU) that conditions a supply air stream, such as environmental air and/or a return air flow from the building 10. After the HVAC unit 12 conditions the air, the air is supplied to the building 10 via ductwork 14 extending throughout the building 10 from the HVAC unit 12. For example, the ductwork 14 may extend to various individual floors or one or more zones (101, 102, 103) of the building 10 and each zone may further comprise one or more outdoor air hoods equipped with filters. In certain embodiments, the HVAC unit 12 may be a heat pump that provides both heating and cooling to the building with one refrigeration circuit configured to operate in different modes. In other embodiments, the HVAC unit 12 may include one or more refrigeration circuits for cooling an air stream and a furnace for heating the air stream.

A control device 16, one type of which may be a thermostat, may be used to designate the temperature of the conditioned air. The control device 16 also may be used to control the flow of air through the ductwork 14. For example, the control device 16 may be used to regulate operation of one or more components of the HVAC unit 12 or other components, such as dampers and fans, within the building 10 that may control flow of air through and/or from the ductwork 14. In some embodiments, other devices may be included in the system, such as pressure and/or temperature transducers or switches that sense the temperatures and pressures of the supply air, return air, and so forth. Moreover, the control device 16 may include computer systems that are integrated with or separate from other building control or monitoring systems, and even systems that are remote from the building 10.

FIG. 2 is a perspective view of an embodiment of the HVAC unit 12. In the illustrated embodiment, the HVAC unit 12 is a single package unit that may include one or more independent refrigeration circuits and components that are tested, charged, wired, piped, and ready for installation. The HVAC unit 12 may provide a variety of heating and/or cooling functions, such as cooling only, heating only, cooling with electric heat, cooling with dehumidification, cooling with gas heat, or cooling with a heat pump. As described above, the HVAC unit 12 may directly cool and/or heat an air stream provided to the building 10 to condition a space in the building 10.

As shown in the illustrated embodiment of FIG. 2, a cabinet 24 encloses the HVAC unit 12 and provides structural support and protection to the internal components from environmental and other contaminants. In some embodiments, the cabinet 24 may be constructed of galvanized steel and insulated with aluminum foil faced insulation. Rails 26 may be joined to the bottom perimeter of the cabinet 24 and provide a foundation for the HVAC unit 12. In certain embodiments, the rails 26 may provide access for a forklift and/or overhead rigging to facilitate installation and/or removal of the HVAC unit 12. In some embodiments, the rails 26 may fit onto "curbs" on the roof to enable the HVAC

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unit 12 to provide air to the ductwork 14 from the bottom of the HVAC unit 12 while blocking elements such as rain from leaking into the building 10.

The HVAC unit 12 includes heat exchangers 28 and 30 in fluid communication with one or more refrigeration circuits. Tubes within the heat exchangers 28 and 30 may circulate refrigerant, such as R-410A, through the heat exchangers 28 and 30. The tubes may be of various types, such as multi-channel tubes, conventional copper or aluminum tubing, and so forth. Together, the heat exchangers 28 and 30 may implement a thermal cycle in which the refrigerant undergoes phase changes and/or temperature changes as it flows through the heat exchangers 28 and 30 to produce heated and/or cooled air. For example, the heat exchanger 28 may function as a condenser where heat is released from the refrigerant to ambient air, and the heat exchanger 30 may function as an evaporator where the refrigerant absorbs heat to cool an air stream. In other embodiments, the HVAC unit 12 may operate in a heat pump mode where the roles of the heat exchangers 28 and 30 may be reversed. That is, the heat exchanger 28 may function as an evaporator and the heat exchanger 30 may function as a condenser. In further embodiments, the HVAC unit 12 may include a furnace for heating the air stream that is supplied to the building 10. While the illustrated embodiment of FIG. 2 shows the HVAC unit 12 having two of the heat exchangers 28 and 30, in other embodiments, the HVAC unit 12 may include one heat exchanger or more than two heat exchangers.

The heat exchanger 30 is located within a compartment 31 that separates the heat exchanger 30 from the heat exchanger 28. Fans 32 draw air from the environment through the heat exchanger 28. Air may be heated and/or cooled as the air flows through the heat exchanger 28 before being released back to the environment surrounding the HVAC unit 12. A blower assembly 34, powered by a motor 36, draws air through the heat exchanger 30 to heat or cool the air. The heated or cooled air may be directed to the building 10 by the ductwork 14, which may be connected to the HVAC unit 12. Before flowing through the heat exchanger 30, the conditioned air flows through one or more filters 38 that may remove particulates and contaminants from the air. In certain embodiments, the filters 38 may be disposed on the air intake side of the heat exchanger 30 to prevent contaminants from contacting the heat exchanger 30.

The HVAC unit 12 also may include other equipment for implementing the thermal cycle. Compressors 42 increase the pressure and temperature of the refrigerant before the refrigerant enters the heat exchanger 28. The compressors 42 may be any suitable type of compressors, such as scroll compressors, rotary compressors, screw compressors, or reciprocating compressors. In some embodiments, the compressors 42 may include a pair of hermetic direct drive compressors arranged in a dual stage configuration 44. However, in other embodiments, any number of the compressors 42 may be provided to achieve various stages of heating and/or cooling. Additional equipment and devices may be included in the HVAC unit 12, such as a solid-core filter drier, a drain pan, a disconnect switch, an economizer, pressure switches, phase monitors, and humidity sensors, among other things.

The HVAC unit 12 may receive power through a terminal block 46. For example, a high voltage power source may be connected to the terminal block 46 to power the equipment. The operation of the HVAC unit 12 may be governed or regulated by a control board 48. The control board 48 may include control circuitry connected to a thermostat, sensors, and alarms. One or more of these components may be

referred to herein separately or collectively as the control device 16. The control circuitry may be configured to control operation of the equipment, provide alarms, and monitor safety switches. Wiring 49 may connect the control board 48 and the terminal block 46 to the equipment of the HVAC unit 12.

FIG. 3 illustrates a residential heating and cooling system 50, also in accordance with present techniques. The residential heating and cooling system 50 may provide heated and cooled air to a residential structure, as well as provide outside air for ventilation and provide improved indoor air quality (IAQ) through devices such as ultraviolet lights and air filters. In the illustrated embodiment, the residential heating and cooling system 50 is a split HVAC system. In general, a residence 52 conditioned by a split HVAC system may include refrigerant conduits 54 that operatively couple the indoor unit 56 to the outdoor unit 58. The indoor unit 56 may be positioned in a utility room, an attic, a basement, and so forth. The outdoor unit 58 is typically situated adjacent to a side of residence 52 and is covered by a shroud to protect the system components and to prevent leaves and other debris or contaminants from entering the unit. The refrigerant conduits 54 transfer refrigerant between the indoor unit 56 and the outdoor unit 58, typically transferring primarily liquid refrigerant in one direction and primarily vaporized refrigerant in an opposite direction.

When the system shown in FIG. 3 is operating as an air conditioner, a heat exchanger 60 in the outdoor unit 58 serves as a condenser for re-condensing vaporized refrigerant flowing from the indoor unit 56 to the outdoor unit 58 via one of the refrigerant conduits 54. In these applications, a heat exchanger 62 of the indoor unit functions as an evaporator. Specifically, the heat exchanger 62 receives liquid refrigerant, which may be expanded by an expansion device, and evaporates the refrigerant before returning it to the outdoor unit 58.

The outdoor unit 58 draws environmental air through the heat exchanger 60 using a fan 64 and expels the air above the outdoor unit 58. When operating as an air conditioner, the air is heated by the heat exchanger 60 within the outdoor unit 58 and exits the unit at a temperature higher than it entered. The indoor unit 56 includes a blower or fan 66 that directs air through or across the indoor heat exchanger 62, where the air is cooled when the system is operating in air conditioning mode. Thereafter, the air is passed through ductwork 68 that directs the air to the residence 52. The overall system operates to maintain a desired temperature as set by a system controller. When the temperature sensed inside the residence 52 is higher than the set point on the thermostat, or the set point plus a small amount, the residential heating and cooling system 50 may become operative to refrigerate additional air for circulation through the residence 52. When the temperature reaches the set point, or the set point minus a small amount, the residential heating and cooling system 50 may stop the refrigeration cycle temporarily.

The residential heating and cooling system 50 may also operate as a heat pump. When operating as a heat pump, the roles of heat exchangers 60 and 62 are reversed. That is, the heat exchanger 60 of the outdoor unit 58 will serve as an evaporator to evaporate refrigerant and thereby cool air entering the outdoor unit 58 as the air passes over the outdoor heat exchanger 60. The indoor heat exchanger 62 will receive a stream of air blown over it and will heat the air by condensing the refrigerant.

In some embodiments, the indoor unit 56 may include a furnace system 70. For example, the indoor unit 56 may

include the furnace system 70 when the residential heating and cooling system 50 is not configured to operate as a heat pump. The furnace system 70 may include a burner assembly and heat exchanger, among other components, inside the indoor unit 56. Fuel is provided to the burner assembly of the furnace system 70 where it is mixed with air and combusted to form combustion products. The combustion products may pass through tubes or piping in a heat exchanger, separate from heat exchanger 62, such that air directed by the blower or fan 66 passes over the tubes or pipes and extracts heat from the combustion products. The heated air may then be routed from the furnace system 70 to the ductwork 68 for heating the residence 52.

FIG. 4 is an embodiment of a vapor compression system 72 that can be used in any of the systems described above. The vapor compression system 72 may circulate a refrigerant through a circuit starting with a compressor 74. The circuit may also include a condenser 76, an expansion valve(s) or device(s) 78, and an evaporator 80. The vapor compression system 72 may further include a control panel 82 that has an analog to digital (A/D) converter 84, a microprocessor 86, a non-volatile memory 88, and/or an interface board 90. The control panel 82 and its components may function to regulate operation of the vapor compression system 72 based on feedback from an operator, from sensors of the vapor compression system 72 that detect operating conditions, and so forth.

In some embodiments, the vapor compression system 72 may use one or more of a variable speed drive (VSDs) 92, a motor 94, the compressor 74, the condenser 76, the expansion valve or device 78, and/or the evaporator 80. The motor 94 may drive the compressor 74 and may be powered by the variable speed drive (VSD) 92. The VSD 92 receives alternating current (AC) power having a particular fixed line voltage and fixed line frequency from an AC power source, and provides power having a variable voltage and frequency to the motor 94. In other embodiments, the motor 94 may be powered directly from an AC or direct current (DC) power source. The motor 94 may include any type of electric motor that can be powered by a VSD or directly from an AC or DC power source, such as a switched reluctance motor, an induction motor, an electronically commutated permanent magnet motor, or another suitable motor.

The compressor 74 compresses a refrigerant vapor and delivers the vapor to the condenser 76 through a discharge passage. In some embodiments, the compressor 74 may be a centrifugal compressor. The refrigerant vapor delivered by the compressor 74 to the condenser 76 may transfer heat to a fluid passing across the condenser 76, such as ambient or environmental air 96. The refrigerant vapor may condense to a refrigerant liquid in the condenser 76 as a result of thermal heat transfer with the environmental air 96. The liquid refrigerant from the condenser 76 may flow through the expansion device 78 to the evaporator 80.

The liquid refrigerant delivered to the evaporator 80 may absorb heat from another air stream, such as a supply air stream 98 provided to the building 10 or the residence 52. For example, the supply air stream 98 may include ambient or environmental air, return air from a building, or a combination of the two. The liquid refrigerant in the evaporator 80 may undergo a phase change from the liquid refrigerant to a refrigerant vapor. In this manner, the evaporator 80 may reduce the temperature of the supply air stream 98 via thermal heat transfer with the refrigerant. Thereafter, the vapor refrigerant exits the evaporator 80 and returns to the compressor 74 by a suction line to complete the cycle.

In some embodiments, the vapor compression system 72 may further include a reheat coil in addition to the evaporator 80. For example, the reheat coil may be positioned downstream of the evaporator relative to the supply air stream 98 and may reheat the supply air stream 98 when the supply air stream 98 is overcooled to remove humidity from the supply air stream 98 before the supply air stream 98 is directed to the building 10 or the residence 52.

It should be appreciated that any of the features described herein may be incorporated with the HVAC unit 12, the residential heating and cooling system 50, or other HVAC systems. Additionally, while the features disclosed herein are described in the context of embodiments that directly heat and cool a supply air stream provided to a building or other load, embodiments of the present disclosure may be applicable to other HVAC systems as well. For example, the features described herein may be applied to mechanical cooling systems, free cooling systems, chiller systems, or other heat pump or refrigeration applications.

As set forth above, embodiments of the present disclosure are directed to a collapsible air hood assembly that is configured to couple to an HVAC unit and transition from a collapsed configuration to a deployed configuration and vice versa. The air hood assembly may be configured to couple to the HVAC unit 12 of FIG. 1, the outdoor unit 58 illustrated in FIG. 3, and/or any other suitable HVAC unit. Additionally, the air hood assembly may transition from the deployed configuration to the collapsed configuration when transportation of the HVAC unit is desired. Therefore, the air hood assembly may remain coupled to the HVAC unit during transportation, thereby reducing or eliminating separate shipping costs for each of the air hood assembly and the HVAC unit. Additionally, adjusting the air hood assembly between the collapsed configuration and the deployed configuration may be less time-consuming when compared to a traditional installation of the air hood assembly. As such, an assembly time of the HVAC unit may also be reduced.

FIG. 5 is a perspective view of a side 102 (e.g., lateral side or surface) of an HVAC unit 100, illustrating an embodiment of an air hood assembly 200 (e.g., hood assembly, intake hood assembly) having an inlet hood 202 in a deployed configuration. As shown in the illustrated embodiment of FIG. 5, the HVAC unit 100 may include a housing 104 configured to support and secure the air hood assembly 200 on the side 102 of the HVAC unit 100. The housing 104 may also support and/or couple to other various components of the HVAC unit 100. For example, the housing 104 may include a first sub-frame 106 (e.g., frame, frame member, support structure) to which the inlet hood 202 may be secured, and a second sub-frame 108 (e.g., frame, frame member, support structure) to which an exhaust hood 204 of the HVAC unit 100 may be secured. The exhaust hood 204 may be configured to protect or cover a louver 205, which may be configured to regulate a flow of exhaust air discharged from the HVAC unit 100. It should be noted that a similar louver may be present within the inlet hood 202 and may be configured to regulate a flow of intake air received by the HVAC unit 100.

The air hood assembly 200 may include a top panel 206 (e.g., top panel assembly, top cover) and a filter assembly 208, each rotatably coupled to the first sub-frame 106. The top panel 206 includes a first panel portion 210 and a second panel portion 212 that may be adjustably coupled to one another. For example, during a transition of the air hood assembly 200 from a deployed configuration to a collapsed configuration, the second panel portion 212 may slide or translate relative to the first panel portion 210 to enable the

first panel portion 210 to substantially cover and contain the filter assembly 208 within and/or against the side 102 of the housing 104 of the HVAC unit 100, as described in greater detail below. The filter assembly 208 may include a frame 213 (e.g., support structure) configured to support filters of the air hood assembly 200. For example, the frame 213 may include a first filter frame 214 (e.g., sub-frame) rotatably coupled to the top panel 206 and a second filter frame 216 (e.g., sub-frame) rotatably coupled to the first sub-frame 106 of the housing 104. The first filter frame 214 and the second filter frame 216 may each be configured to support one or more filters 215, 217. The filters 215, 217 may be configured to block particulates and/or contaminants from entering an interior of the HVAC unit 100 from the environment surrounding the HVAC unit 100, such as during operation of the HVAC unit 100. The filter assembly 208 may also include a mesh 218 to be configured to secure and hold the one or more filters 215, 217 in the first filter frame 214 and the second filter frame 216.

The first filter frame 214 and the second filter frame 216 may also be rotatably coupled to one another by one or more hinge joints 219 (e.g., hinges, pivot connections, etc.). The one or more hinge joints 219 may be configured to couple the first and second filter frames 214, 216 together to facilitate the transition of the air hood assembly 200 from a deployed configuration to a collapsed configuration, as described in greater detail with reference to FIGS. 8 and 11A. As discussed in detail herein, the filter assembly 208 may be positioned over an opening (e.g., air inlet) of the HVAC unit 100. Accordingly, air may be drawn through the filters 215, 217 and into an interior of the HVAC unit 100 via a fan or blower of the HVAC unit 100. The air may be directed through the HVAC unit 100 and may pass over one or more heat exchangers within the HVAC unit 100 to condition the air.

In the deployed configuration, the first filter frame 214 and the second filter frame 216 may be extended and/or expanded out towards the environment and away from the HVAC unit 100 such that the first filter frame 214 and the second filter frame 216 are aligned with one another (e.g., side-by-side, adjacent one another) in an extended position. In the deployed configuration, a gap 222 may be formed between the first filter frame 214 and the second filter frame 216. The gap 222 may be occluded by a flexible material (e.g., a filter and/or cloth strip) configured to span the gap 222 and between the first filter frame 214 and the second filter frame 216. The gap 222 may be formed in the deployed configuration as a result of the spatial relationship of the first filter frame 214 relative to the second filter frame 216 (e.g., via the hinge joints 219), and the spatial relationship may facilitate transition of the air hood assembly 200 from the deployed configuration to the collapsed configuration and vice versa, as described in greater detail below. In the deployed configuration, the air hood assembly 200 may further incorporate a pair of side panels 224 (e.g., on opposite sides of the inlet hood 202) configured to couple to, secure, and/or support the top panel 206 and the filter assembly 208 in the extended position while in the air hood assembly 200 is in the deployed configuration. As shown in the illustrated embodiment of FIG. 5, the side panels 224 may extend laterally outward from the side 102 of the HVAC unit 100. For example, the side panels 224 may each include triangular geometry or profile having a first edge 225 configured to couple to and/or abut the housing 104 of the HVAC unit 100. Additionally, a second edge 226 of each side panel 224 may be configured to couple to the top panel 206. Further still, the side panels 224 may each include a

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third edge 227 configured to couple to and/or abut the filter assembly 208. The third edges 227 of the side panels 224 may also be configured to secure or lock the filter assembly 208 in the extended position while the air hood assembly 200 is in the deployed configuration. As discussed further below, the side panels 224 may be removed from the air hood assembly 200 to facilitate transition of the air hood assembly 200 from the deployed configuration to the collapsed configuration.

FIG. 6 is a perspective view of an embodiment of the air hood assembly 200 in the deployed configuration. The inlet hood 202 is configured to block water and/or other contaminants from entering the interior of the HVAC unit 100 and/or otherwise through the filters 215, 217. As discussed above, the top panel 206 may include the first panel portion 210 and the second panel portion 212 and may be disposed at an oblique angle 150 relative to the side 102 of the HVAC unit 100 such that the top panel 206 may provide protection for components of the air hood assembly 200 positioned below the top panel 206 (e.g., the filter assembly 208) and/or within the housing of the HVAC system 100 (e.g., a louver). The first panel portion 210 may have a top edge 231 (e.g., a base edge, first edge), a bottom edge 232 (e.g., a distal edge, second edge) opposite the top edge 231 and having a lip 300, and a pair of lateral edges 233, 234. The second panel portion 212 may also have a top edge 235 (e.g., overlapping edge, first edge), a bottom edge 236 (e.g., a distal edge, second edge) opposite the top edge 235 and having a lip 302, and a pair of lateral edges 237, 238. As illustrated, the top edge 235 of the second panel portion 212 may be positioned underneath or behind the first panel portion 210 to enable the second panel portion 212 to slide relative to the first panel portion 210. In the collapsed configuration of the air hood assembly 200, the second panel portion 212 may be retracted behind and substantially covered (e.g., via overlap) by the first panel portion 210. The first panel portion 210 may be configured to maintain its shape or geometry while transitioning from a deployed configuration to a collapsed configuration. For example, regardless of configuration (e.g., deployed, transitioning, collapsed) of the air hood assembly 200, the first panel portion 210 may remain fixed relative to the second panel portion 212, and the second panel portion 212 may slide or translate towards or away from the first panel portion 210. That is, the lip 302 of the second panel portion 212 may move towards the lip 300 of the first panel portion 210 when the air hood assembly 200 transitions from the deployed configuration to the collapsed configuration. For example, the second panel portion 212 may translate relative to the first panel portion 210 until the bottom edge 236 and the lip 302 of the second panel portion 212 contact and/or are adjacent to the bottom edge 232 and the lip 300 of the first panel portion 210. In the collapsed configuration, the first panel portion 212 may substantially cover and/or overlap with the second panel portion 210, as illustrated in FIG. 12. When the air hood assembly 200 transitions from the collapsed configuration to the deployed configuration, the bottom edge 236 and the lip 302 of the second panel portion 212 may extend away from the bottom edge 232 and the lip 300 of the first panel portion 210. In this manner, the top panel 206 enables the filter assembly 208 to extend out into the deployed configuration (e.g., with filter frames 214, 216 side-by-side). It should be noted that, as used herein, the term “top” is not limited to orientations or spatial relationships whereby an element (e.g., the top panel 206) is located above (e.g., vertically above) other elements (e.g., relative to gravity). That is, in some configurations (e.g., the collapsed configuration), the top panel 206 may be the most outward

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element relative to the housing 104 such that the top panel 206 substantially covers and contains other components of the air hood assembly 200 within the housing 104, as described in greater detail below.

As illustrated, the first edges 225 of the side panels 224 may be coupled to the housing 104 via one or more fasteners 304. The fasteners 304 may be pins or screws that are readily removed to facilitate the transition of the air hood assembly 200 from the deployed configuration to the collapsed configuration. The second edge 226 of the side panel 224 may be coupled to the top panel 206 via one or more fasteners 305. The fasteners 305 may be similar to fasteners 304 to enable ready removal of the side panels 224 to enable transition of the air hood assembly 200 to the collapsed configuration. The third edge 227 of the side panel 224 may be positioned above the filter assembly 208 (e.g., relative to gravity) in the deployed configuration in order to bias (e.g., via a force 310 in an outward and/or downward direction) and/or retain (e.g., secure) the filter assembly 208 in the extended positioned.

As described above, the filter assembly 208 may include the first filter frame 214, the second filter frame 216, filters 215, 217, and the mesh 218. The first filter frame 214 may have a top side 240 (e.g., distal side, first side), a bottom side 241 (adjoining side, second side), and a pair of lateral sides 242, 243. The second filter frame 216 may also have a top side 244 (e.g., adjoining side, first side), a bottom side 245 (e.g., mounting side, second side), and a pair of lateral sides 246, 247. When the filter assembly 208 is in the extended position, the bottom side 241 of the first filter frame 214 and the top side 244 of the second filter frame 216 may be separated by the gap 222. The bottom side 241 of the first filter frame 214 and the top side 244 of the second filter frame 216 are also rotatably coupled to one another such that the first filter frame 214 and the second filter frame 216 may angularly rotate towards and/or away from one another during transition of the air hood assembly 200 between the deployed configuration and the collapsed configuration via the hinge joints 219. The spatial relationship of the first filter frame 214 and the second filter frame 216 that results in the gap 222 in the deployed configuration may also provide additional space for the first filter frame 214 and the second filter frame 216 to rotate relative to one another. The gap 222 may be enclosed by a flexible material, such as a filter cloth strip, configured to fit within (e.g., fill, span) the gap 222, thereby blocking air flow between the first filter frame 214 and the second filter frame 216 and into the inlet hood 202.

The top side 240 of the first filter frame 214 may be rotatably coupled to the bottom edge 236 of the second panel portion 212. Further, each of the sides 240, 241, 242, 243, 244, 245, 246, and 247 of the first filter frame 214 and the second filter frame 216 may include or be formed from C-channels configured to facilitate installation of the filters 215, 217 therein. It should be noted that each of the filters 215, 217 may each include multiple individual filters. When the filter 215 is installed within the first filter frame 214, a gap (e.g., space, void) may extend between an edge of the filter 215 and the top side 240 of the first filter frame 214. This gap (e.g., within the C-channel forming the top side 240) may provide room for the filter 215 to be adjusted or manipulated during replacement of one or more of the filters 215 (e.g., without disassembling the filter assembly 208), as described in greater detail with reference to FIG. 13. It should be noted that each of the filter frame sides 240, 241, 242, 243, 244, 245, 246, and 247, which may be formed or defined by one or more C-channels, may provide a gap similar to that described above between the side of the

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corresponding filter frame **214**, **216** and a respective side of the corresponding filter **215**, **217** to facilitate installation and/or removal of the filters **215**, **217**.

As discussed above, various components of the air hood assembly **200** may be coupled to the first sub-frame **106**. The first sub-frame **106** may have a top portion **111** (e.g., first portion), a bottom portion **112** (e.g., second portion) opposite the top portion **111** relative to an air inlet protected by the air hood assembly **200**, a first side portion **113**, and a second side portion **114** opposite the first side portion **113** relative to the air inlet protected by the air hood assembly **200**. The top portion **111** may also include a top panel retainer **115** (e.g., clasp, clamp, bracket), which is described further below. Each portion **111**, **112**, **113**, and **114** of the first sub-frame **106** may be configured to couple to and secure components of the air hood assembly **200** to the housing **104** of the HVAC unit **100**. For example, the bottom side **245** of the second filter frame **216** may be rotatably coupled to the bottom portion **112** of the first sub-frame **106** via hinge joints **220**, and the top edge **231** of the first panel portion **210** may be rotatably coupled to the first portion **111** of the first sub-frame **106** via the top panel retainer **115**. Further, while in the deployed configuration, the first edges **225** of the side panels **224** may be coupled to the first side portion **113** and the second side portion **114**, respectively, via the fasteners **304**.

FIG. 7A is a side perspective view of an embodiment of the top panel retainer **115**, which may be configured to facilitate the securement of the top edge **231** of the first panel portion **210** to the housing **104** of the HVAC system **100** in the deployed configuration. For example, the top panel retainer **115** may be secured to the top portion **111** of the first sub frame **106** via one or more fasteners **306**. The one or more fasteners **306** may extend through the top portion **111** of the first sub frame **106** and into the top panel retainer **115**, thereby securing the top panel retainer **115** to the housing **104**. The top edge **231** of the first panel portion **210** may have a lip **310** that extends in a generally vertical direction when the air hood assembly **200** is in the deployed configuration, and the top panel retainer **115** may also have a lip **116** that extends in a generally horizontal direction.

As illustrated and discussed above, fasteners **304** may extend through the first edge **225** of the side panel **224** and into the second side portion **114** of the first sub-frame **106** to couple and secure the side panel **224** to the housing **104**. Further, fasteners **305** may extend through the first panel portion **210** and into the second edge **226** of the side panel **224**, thereby coupling and securing the first panel portion **210** to the side panel **224**. When the fasteners **304**, **305** are removed along with the side panels **224** (e.g., to transition the air hood assembly **200** from the deployed configuration to the collapsed configuration), the top panel **206** may rotate or pivot towards the housing **104** in a generally downward and/or inward direction **400** relative to the housing **104**, and the lip **310** may rotate or pivot towards the lip **116** of the top panel retainer **115** in a generally downward and/or inward direction **350** relative to the lip **116**. In the collapsed configuration, the top panel **206** extends along and generally flush with the housing **104**, as discussed above. Additionally, in this position of the top panel **206**, the lip **116** may extend along and rest against (e.g., above and/or on top of, relative to gravity) the lip **116**, thereby securing the top panel **206** within the top panel retainer **115** in the collapsed configuration, as discussed in greater detail below with reference to FIG. 7B.

FIG. 7B is an exploded view of an embodiment of the top panel retainer **115** with the top panel **206** of the air hood

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assembly **200** in the collapsed configuration. When in the collapsed configuration, the lip **310** of the first panel portion **210** may be positioned above the lip **116** of the top panel retainer **115** relative to gravity. As illustrated, the lip **310** of the first panel portion **210** may rest against the lip **116** of the top panel retainer **115** in the collapsed configuration, such that the lip **116** of the top panel retainer **115** supports the top panel **206** via engagement with the lip **310** of the first panel portion **210**. In this way, the top panel retainer **115** restricts or blocks the top panel **206** from moving downward relative to the top panel retainer **115**. For example, the lip **116** of the top panel retainer **115** may be configured to support a weight of the top panel **206** while in the collapsed configuration. Further, the lip **116** may be configured to restrict the top panel **206** from rotating further inward, relative to the housing **104**, upon reaching the collapsed configuration. Furthermore, as shown, the lip **310** and the lip **116** are dimensioned such that the lips **310**, **116** substantially overlap with and contact one another in the collapsed configuration of the air hood assembly **200**. For example, a full length or a substantially full length of the lip **310** rests on the lip **116**, and the lip **116** extends toward the housing **104** beneath (e.g., relative to gravity) the full length or substantially full length of the lip **310**. Thus, the top panel retainer **115** may block or mitigate inadvertent dislodging of the top panel **206** from the top panel retainer **115**.

Turning now to FIG. 8A, a schematic view illustrating transition of an embodiment of the air hood assembly **200** between a deployed configuration and a collapsed configuration, is shown. During transition, the side panels **224** may be removed to enable rotation and/or pivoting of the top panel **206** relative to the housing **104**, as enabled by the lip **116** of the top panel retainer **115** and the lip **310** of the first panel portion **210**. As discussed above, the top panel **206** may include the first panel portion **210** and the second panel portion **212**. During transition from the deployed configuration to the collapsed configuration, the top panel **206** may rotate in the downward and/or inward direction **400** relative to the housing **104** such that the filter assembly **208** may collapse and be protected and contained within the housing **104**. Further, during transition, the second panel portion **212** may move in a generally inward direction **402** relative to the first panel portion **210**. Because the second panel portion **212** is coupled to the first filter frame **214**, when the second panel portion **212** slides inward relative to the first panel portion **210**, the first filter frame **214** also moves inward towards the housing **104** of the HVAC unit **100** such that when the air hood assembly **200** is fully collapsed, the filter assembly **208** is covered and contained within the housing **104** by the top panel **206**. During transition from the collapsed configuration to the deployed configuration, the top panel **206** may rotate in an outward and/or upward direction **500** relative to the housing **104**. Further, the second panel portion **212** may move in a generally outward direction **502** relative to the first panel portion **210**. In turn, the first filter frame **214** may move away from the housing **104** to provide additional space for the filter assembly **208** to move (e.g., rotate and/or pivot) into the extended position. To facilitate transition, the first filter frame **214** and the second filter frame **216** may be coupled together via the one or more hinge joints **219**. The one or more hinge joints **219** may be configured to enable rotational movement between the first filter frame **214** and the second filter frame **216**. Further, one or more additional hinge joints **220** may couple the second filter frame **216** to the first sub-frame **106**. The one or more additional hinge joints **220** may enable rotational movement between the bottom side **245** of the second

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filter frame 216 and the bottom portion 112 of the first sub-frame 106. Such features are described in greater detail with reference to FIG. 11A and FIG. 11B.

As discussed above, the first filter frame 214 and the second filter frame 216 are rotatably coupled to one another by the one or more hinge joints 219. During transition from the deployed configuration to the collapsed configuration, the first filter frame 214 and the second filter frame 216 may rotate towards each other (e.g., about the one or more hinge joints 219) until the air hood assembly 200 is in the collapsed configuration. When the air hood assembly 200 is fully collapsed, the first filter frame 214 and the second filter frame 216 overlap with one another (e.g., in a side-by-side or adjacent arrangement) and may be substantially contained against the housing 104 (e.g., covered) by the top panel 206. For example, the bottom side 241 of the first filter frame 214 and the top side 244 of the second filter frame 216 may be rotatably coupled to one another via the hinge joints 219 such that the bottom side 241 of the first filter frame 214 and the top side 244 of the second filter frame 216 may move in a generally inward and/or upward direction 404 relative to the housing 104 when the air hood assembly 200 transitions from the deployed configuration to the collapsed configuration. As the bottom side 241 of the first filter frame 214 and the top side 244 of the second filter frame 216 pivot or rotate and move in the inward and/or upward direction 404, an inward surface 270 of the first filter frame 214 may pivot an inward and/or upward direction 274 relative to the top panel 206. Further, an inward surface 280 of the second filter frame 216 may pivot in an inward and/or upward direction 284 relative to the first sub-frame 106 of the housing 104. Thus, during transition from the deployed configuration to the collapsed configuration, the first filter frame 214 and the second filter frame 216 may rotate or pivot towards one another until they are substantially flush against one another and arranged to overlap one another. For example, in the collapsed configuration and relative to the housing 104, the first panel portion 210 is external to the second panel portion 212, the second panel portion 212 is external to the first filter frame 214, and the first filter frame 214 is external to the second filter frame 216. Thus, the first panel portion 210 substantially covers and contains the second panel portion 210 and the first and second filter frames 214, 216 against the housing 104 in the collapsed configuration. It should be noted that, as the second panel portion 212 moves in the inward direction 402 relative to the first panel portion 210, the top side 240 of the first filter frame 214 may move in a similar direction due to the coupling between the second panel portion 212 and the first filter frame 214.

During transition from the collapsed configuration to the deployed configuration, the bottom side 241 of the first filter frame 214 and the top side 244 of the second filter frame 216 may move in a generally outward and/or downward direction 504 relative to the housing 104 to enable transition of the filter assembly 208 into the extended position associated with the deployed configuration of the air hood assembly 200. As the bottom side 241 of the first filter frame 214 and the top side 244 of the second filter frame 216 move in the outward and/or downward direction 504 relative to the housing 104, an outward surface 272 of the first filter frame 214 may rotate in an outward and/or downward direction 276 relative to the top panel 206 and an outward surface 282 of the second filter frame 216 may rotate in an outward and/or downward direction 286 relative to the first sub frame 106 of the housing 104. Thus, during transition from the collapsed configuration to the deployed configuration, the first filter frame 214 and the second filter frame 216 may

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rotate away from one another such that, when extended, the first filter frame 214 and the second filter frame 216 are aligned with one another (e.g., end-to-end) in the extended position. It should be noted that, as the second panel portion 212 moves outward relative to the first panel portion 210, the top side 240 of the first filter frame 214 may move in a similar direction due to the coupling between the second panel portion 212 and the first filter frame 214.

FIG. 8B a perspective view of an embodiment of the air hood assembly 200 transitioning from a deployed configuration to a collapsed configuration. As described above, the filter assembly 208 may be configured to fold or collapse towards the housing 104 when transitioning from the deployed configuration to the collapsed configuration. In the illustrated embodiment, as the filter assembly 208 collapses towards the first sub-frame 106 of the housing 104, the first filter frame 214 may be configured to sit generally flush against the top panel 206 such that the top panel 206 may cover (e.g., be disposed external to, relative to the housing 104) a portion of the first filter frame 214 and the filter assembly 208. Further, because the bottom edge 236 of the second panel portion 212 is rotatably coupled to the top side 240 of the first filter frame 214, as the second panel portion 212 slides or translates relative to the first panel portion 210, the first filter frame 214 may also slide or translate relative to the first panel portion 210 towards the housing 104. Thus, when in the collapsed configuration, the first panel portion 210 substantially covers and contains the filter assembly 208 within and/or against the housing 104.

FIG. 9 is a top-perspective view of an embodiment of the air hood assembly 200 in a deployed configuration. In the deployed configuration, the side panels 224 may be coupled to the first sub-frame 106 via fasteners 304 and to the first panel portion 210 via fasteners 305. Once the fasteners 304, 305 and side panels 224 are removed, the top panel 206 may rotate towards the first sub-frame 106 to the collapsed configuration. As discussed above, the top edge 235 of the second panel portion 212 may be positioned underneath the bottom edge 232 of the first panel portion 210, and the second panel portion 212 may be configured to slide or translate relative to the first panel portion 210 to enable the air hood assembly 200 to transition to the collapsed configuration. To facilitate this movement, the second panel portion 212 may have a slot 320 formed in each of the lateral edges 237, 238 of the second panel portion 212. The slot 320 may be configured to guide or direct the second panel portion 212 towards the first panel portion 210 via one or more bolts, as described in greater detail below.

For example, FIG. 10A shows an exploded view of an embodiment of the slot 320 configured to facilitate movement of the second panel portion 212 relative to the first panel portion 210. The slot 320 may span across a length 322 of the lateral edge 237 of the second panel portion 212. As discussed above, the top edge 235 of the second panel portion 212 may be positioned underneath the bottom edge 232 of the first panel portion 210, thereby enabling the second panel portion 212 to translate in either the inward direction 402 or outward direction 502 relative to the first panel portion 210, such as along a first axis 500. To facilitate the sliding of the second panel portion 212 relative to the first panel portion 210, a first bolt 330 (e.g., pin, fastener, etc.) may extend through the first panel portion 210 at the bottom edge 232 of the first panel portion 210 and into the slot 320. The first bolt 330 may be configured to couple the first panel portion 210 to the second panel portion 212 while also enabling movement between the first panel portion 210 and the second panel portion 212. That is, the first bolt 330

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may be tight enough to secure the first panel portion 210 to the second panel portion 212, but loose enough to enable movement between the two portions 210, 212. A second bolt 332 (e.g., pin, fastener, etc.) may be positioned within the slot 320 along or at the bottom edge 236 of the second panel portion 212. One or more fasteners 308 may be configured to secure the second panel portion 212 to the side panel 224 while the air hood assembly 200 is in the deployed configuration. When transitioning from the deployed configuration to the collapsed configuration, the fasteners 305, 308 and the side panels 224 may be removed, and the first bolt 330 and the second bolt 332 may move along the slot 320 towards the first bolt 330 within the slot 320 to guide the second panel portion 212 in the upward direction 402 along the first axis 500. For example, because the first panel portion 210 remains fixed relative to the second panel portion 212, the first bolt 330 may also remain in a fixed position along the bottom edge 232 of the first panel portion 210. As the second panel portion 212 slides towards the first panel portion 210 along the first axis 500, the first bolt 330 may be contained within the slot 320 thereby limiting movement along a second axis 510. It should be noted that a similar (e.g., additional) slot 320 may be formed along the lateral edge 238 of the second panel portion 212 and may be configured to function similar to the slot 320 described above.

FIG. 10B is an exploded view of an embodiment of the coupling between the first panel portion 210 and the second panel portion 212 via the first bolt 330. As discussed above, the fastener 305 may be configured to couple the first panel portion 210 to the side panel 224 and may be readily removed to enable the second panel portion 210 to slide relative to the first panel portion 210 when the air hood assembly 200 transitions to the collapsed configuration. As illustrated, the bottom edge 232 of the first panel portion 210 is positioned above and/or external to the top edge 235 of the second panel portion 212, and the first bolt 330 may be configured to extend through the bottom edge 232 of the first panel portion 210 and into the second panel portion 212 via the slot 320. The first bolt 330 may be retained within the slot 320 by a nut 331 that is configured to secure the coupling between the first panel portion 210 and the second panel portion 212, as described above. That is, the nut 331 may be tight enough to couple the first panel portion 210 to the second panel portion 212 to one another, but loose enough to enable movement of the second panel portion 212 relative to the first panel portion 210 along the axis 500. For example, the first bolt 330 may extend through and be contained within the slot 320 such that as the second panel portion 212 may slide relative to the first panel portion 210, and movement (e.g., lateral movement) that is not along the axis 500 may be limited by the first bolt 330 retained within the slot 320. It should be noted that the lateral edge 238 of the second panel portion 212 may have a similar slot and fastener configuration as that discussed above.

As discussed above, one or more fasteners may be used to rotatably couple various components of the air hood assembly 200 to one another. FIG. 11A is a perspective view of an embodiment of the hinge joint 219 configured to rotatably couple the first filter frame 214 and the second filter frame 216. As discussed above, the gap 222 may be formed between the first filter frame 214 and the second filter frame 216 to provide additional space for the filter frames 214, 216 to rotate relative to one another. The hinge joint 219 may include a joining plate 600 (e.g., a hinge plate) including a first hole 601 and a second hole 602. The first filter frame 214 may be rotatably coupled to the joining plate 600 via a first rivet 603 (e.g., pin) extending through the first hole 601

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and through the lateral side 242 of the first filter frame 214. The second filter frame 216 may be rotatably coupled to the joining plate 600 via a second rivet 604 (e.g., pin) extending through the second hole 602 and through the lateral side 246 of the second filter frame 216. By coupling the first filter frame 214 and the second filter frame 216 to the joining plate 600, the filter frames 214, 216 may rotate relative to each other when the air hood assembly 200 transitions between the deployed configuration and the collapsed configuration. It should be noted that the lateral side 243 of the first filter frame 214 and the lateral side 247 of the second filter frame 216 (illustrated in FIG. 6) may be coupled to one another by a similar hinge joint 219. Further, in some embodiments, the rivets 603, 604 may be self-tapping fastening screws, pins, or other suitable element about which the first filter frame 214 and/or second filter frame 216 may rotate.

FIG. 11B is a perspective view of an embodiment of the hinge joint 220 configured to rotatably couple the second filter frame 216 to the first sub-frame 106. The hinge joint 220 may be an L-shaped bracket (e.g., mounting flange) having a first attaching member 610 (e.g., first flange) and a second attaching member 612 (e.g., second flange). The first attaching member 610 may be configured to secure the hinge joint 220 to the first sub-frame 106 of the housing 104 via one or more bolts 620 or other mechanical fasteners. The one or more bolts 620 may secure the first attaching member 610 to the first sub-frame 106 such that the hinge joint 220 is fixedly attached to the first sub-frame 106. The second attaching member 612 may have a hole 614 configured to facilitate the adjustable coupling between the second filter frame 216 and the first sub-frame 106. That is, the second filter frame 214 may be rotatably coupled to the second attaching member 612 via a rivet 615 extending through the hole 614 and through the lateral side 246 of the second filter frame 216. By coupling the second filter frame 216 to the second attaching member 612, the second filter frame 214 is configured to rotate relative to the first sub-frame 106 of the housing 104, thereby facilitating transition of the air hood assembly 200 between the deployed configuration and the collapsed configuration. It should be noted that a similar hinge joint 220 may be included on the lateral side 247 (illustrated in FIG. 6) of the second filter frame 216, and in some embodiments, the rivet 615 may include a self-tapping fastening screw, pin, or other suitable element about which the second filter frame 216 may rotate.

Referring now to FIG. 12, a perspective view of an embodiment of the air hood assembly 200 in a collapsed configuration is illustrated. As shown, the side panels 224 are removed such that the top panel 206 may rotate to rest along the housing 104 (e.g., in a generally vertical orientation) and to substantially cover and contain the filter assembly 208 against and/or within the housing 104 of the HVAC unit 100. As discussed above, during transition from the deployed configuration to the collapsed configuration, the second panel portion 212 may slide or translate relative to the first panel portion 210, such that the second panel portion 212 may be substantially covered by the first panel portion 210. Indeed, the first panel portion 210 may sit generally flush against the housing 104, thereby protecting and containing the filter assembly 208 within the housing 104. That is, the first panel portion 210 may overlap and be disposed external to the second panel portion 212 and the filter assembly 208 contained within the housing 104. For example, as illustrated, the bottom edge 232 of the first panel portion 210 is substantially aligned with the bottom edge 236 of the second panel portion 212, such that the second panel portion 212 is substantially covered and/or disposed

behind by the first panel portion **210** (e.g., relative to an external facing surface of the first panel portion **210**). That is, in the collapsed configuration, the first panel portion **210** may be an outermost layer of the air hood assembly **200** relative to the housing **104**, and the second panel portion **212** may be positioned internal to the first panel portion **210** relative to the housing **104**. Such a configuration enables the top panel **206** to cover and protect the filter assembly **208** within the housing **104** and further aids in installation of the air hood assembly **200**. More specifically, the filters **215**, **217** and the air hood assembly **200** may be installed with the HVAC unit **100** prior to transportation of the HVAC unit **100** (e.g., to an installation site), and the HVAC unit **100** may then be transported (e.g., shipped) without removing the filters **215**, **217** and the air hood assembly **200** because the filter assembly **208** is substantially covered and protected by the top panel **206**. Additionally, in the collapsed configuration, the air hood assembly **200** does not significantly and/or materially increase a footprint of the HVAC unit **100**, thereby reducing delivery, transportation, and installation costs associated with the HVAC unit **100**.

FIG. 13 is a schematic view of an embodiment of the filter assembly **208** with the first filter frame **214** and the second filter frame **216** arranged in the extended position (e.g., end-to-end, adjacent one another) and with the filters **215**, **217** installed. As discussed above, the first filter frame **214** and the second filter frame **216** may be separated by the gap **222** and may be coupled together via the hinge joints **219**. The filters **215**, **217** may be retained within the first and second filter frames **214**, **216** by the mesh **218**. The gap **222** may be formed in the deployed configuration as a result of the spatial relationship of the first filter frame **214** relative to the second filter frame **216** (e.g., via the hinge joints **219**), and the spatial relationship may facilitate transition of the air hood assembly **200** from the deployed configuration to the collapsed configuration and vice versa. Further, each of the sides **240**, **241**, **242**, **243**, **244**, **234**, **246**, and **247** of the filter frames **214**, **216** may include or be formed from C-channels configured to facilitate installation of the filters **215**, **217** therein. For example, when the filters **215**, **217** are installed, a first gap **700** may be formed between a top edge **800** of the filter **215** and the top side **240** of the first filter frame **214**, and a second gap **702** may be formed between a bottom edge **801** of the filter **215** and the bottom side **241** of the first filter frame **214**. Further, a third gap **704** may be formed between a first lateral edge **802** of the filter **215** and the lateral side **242** of the first filter frame **214**, and a fourth gap **706** may be formed between a second lateral edge **803** of the filter **215** and the lateral side **243** of the first filter frame **214**. Similarly, a first gap **710** may be formed between a top edge **804** of the filter **217** and the top side **244** of the second filter frame, and a second gap **712** may be formed between a bottom edge **805** of the filter **217** and the bottom side **245** of the second filter frame **216**. Further, a third gap **714** may be formed between a first lateral edge **806** of the filter **217** and the lateral side **246** of the second filter frame **216**, and a fourth gap **716** may be formed between a second lateral edge **807** of the filter **217** and the lateral side **247** of the second filter frame **216**. Each of the gaps **700**, **702**, **704**, **706**, **710**, **712**, **714**, **716** may be configured to enable the filters **215**, **217** to move relative to the first and second filter frames **214**, **216** and may provide adequate clearance for the filters **215**, **217** to be readily removed or installed as need. For example, to remove the filter **215**, the top edge **800** of the filter **215** may be transitioned into the gap **700** towards the top side **240** of the first filter frame **214** such that the bottom edge **801** of the filter **215** may no longer constrained or held by the bottom

side **241** of the first filter frame **214**, thereby enabling the removal of the filter **215** from the first filter frame **214**. Similarly, to remove the filter **217**, the top edge **804** of the filter **217** may be transitioned into the gap **710** towards the top side **244** of the second filter frame **216** such that the bottom edge **805** of the filter **217** may no longer constrained or held by the bottom side **245** of the second filter frame **216**. It should be noted that any of the gaps **700**, **702**, **704**, **706**, **710**, **712**, **714**, **716** may provide clearance for the filters **215**, **217** to be removed from the filter frames **214**, **216**, and that the removal of filters is not limited to the method described above. That is, in some embodiments, the filters **215**, **217** may be transitioned into the gaps **702**, **712** towards the bottom sides **241**, **245** of the first and second filter frames **214**, **216** such that the top edges **800**, **804** of the filters **215**, **217** are no longer held or constrained by the top sides **240**, **244** of the first and second filter frames **214**, **216**. In other embodiments, the filters **215**, **217** may be moved into the gaps **704**, **706**, **714**, **716** towards the lateral sides **242**, **243**, **246**, **247** of the first and second filter frames **214**, **216** in a similar fashion such that the filters **215**, **217** are no longer constrained by one or more surfaces of the first and second filter frames **214**, **216**. As noted above, each filter **215**, **217** may include one or more individual filters (e.g., a plurality of filters) that each may be individually removed via movement into one of the gaps as described above.

Providing a collapsible air hood assembly capable of transitioning between a deployed configuration and a collapsed configuration, in accordance with the present disclosure, can reduce the size (e.g., footprint) of the HVAC unit and protect components within the air hood assembly (e.g., filters) during transportation, thereby reducing time and transportation costs associated with the delivery of the HVAC units. Further, by configuring the air hood assembly as described above, the air hood assembly may provide protection to certain components of the HVAC system that would otherwise need to be removed and reinstalled upon delivery of the HVAC unit, and additional time and costs associated with assembly of the HVAC unit may be reduced.

The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [performing] [a function] . . .” or “step for [performing] [a function] . . .”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

While only certain features and embodiments of the disclosure have been illustrated and described, many modifications and changes may occur to those skilled in the art, such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, including temperatures and pressures, mounting arrangements, use of materials, colors, orientations, and so forth without materially departing from the novel teachings and advantages of the subject matter recited in the claims. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure. Furthermore, in an effort to provide a concise description of the exemplary embodiments, all features of an actual imple-

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mentation may not have been described, such as those unrelated to the presently contemplated best mode of carrying out the disclosure, or those unrelated to enabling the claimed disclosure. It should be noted that in the development of any such actual implementation, as in any engineering or design project, numerous implementation specific decisions may be made. Such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure, without undue experimentation.

The invention claimed is:

1. A filter assembly for a heating, ventilation, and air conditioning (HVAC) unit, comprising:

- a filter frame configured to adjustably couple to a housing of the HVAC unit and configured to support a plurality of filters, wherein the filter frame comprises:
 - a first sub-frame configured to support a first filter of the plurality of filters;
 - a second sub-frame configured to support a second filter of the plurality of filters;
 - a gap disposed between the first sub-frame and the second sub-frame; and
 - a flexible material extending across the gap, wherein the flexible material is configured to block air flow between the first sub-frame and the second sub-frame,

wherein the filter assembly is configured to adjust between a collapsed configuration and a deployed configuration, and the filter assembly is configured to be contained within the housing in the collapsed configuration.

2. The filter assembly of claim 1, wherein the first sub-frame and the second sub-frame are rotatably coupled to one another.

3. The filter assembly of claim 2, wherein the first sub-frame and the second sub-frame are rotatably coupled to one another via one or more hinge joints such that the first sub-frame and the second sub-frame rotate toward one another and about the one or more hinge joints during transition of the filter assembly from the deployed configuration to the collapsed configuration.

4. The filter assembly of claim 3, wherein the first sub-frame and the second sub-frame rotate away from one another and about the one or more hinge joints during transition of the filter assembly from the collapsed configuration to the deployed configuration.

5. The filter assembly of claim 1, wherein the first sub-frame and the second sub-frame are configured to overlap with one another in the collapsed configuration.

6. The filter assembly of claim 1, wherein the first sub-frame and the second sub-frame are configured to align with one another in a direction extending from the housing in the deployed configuration.

7. The filter assembly of claim 1, wherein the gap is configured to facilitate transition of the filter assembly from the collapsed configuration to the deployed configuration.

8. The filter assembly of claim 1, wherein the first sub-frame and the second sub-frame are each formed with a respective plurality of C-channels, and wherein each C-channel in the respective plurality of C-channels defines an additional gap configured to receive at least one filter of the plurality of filters.

9. The filter assembly of claim 8, wherein the additional gap is configured to define a clearance between the corresponding C-channel and the at least one filter to enable installation and removal of the at least one filter.

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10. A hood assembly for a heating, ventilation, and air conditioning (HVAC) unit, comprising:

- a top panel configured to rotatably couple to a housing of the HVAC unit, wherein the top panel comprises a panel portion and a lip extending crosswise from the panel portion; and

- a filter frame configured to rotatably couple to the top panel, wherein the filter frame is configured to support at least one filter,

wherein the hood assembly is adjustable between a collapsed configuration and a deployed configuration, the top panel is configured to retain the filter frame within the housing in the collapsed configuration, the lip is configured to engage with a top panel retainer of the housing in the collapsed configuration, and the lip is configured to be spaced apart from the top panel retainer of the housing in the deployed configuration.

11. The hood assembly of claim 10, wherein the filter frame comprises a first sub-frame configured to support a first filter and a second sub-frame configured to support a second filter, and wherein the first sub-frame and the second sub-frame are rotatably coupled to one another.

12. The hood assembly of claim 11, wherein the first sub-frame is configured to rotatably couple to the top panel, and the second sub-frame is configured to rotatably couple to the housing.

13. The hood assembly of claim 10, wherein the hood assembly is configured to transition between the collapsed configuration and the deployed configuration with the at least one filter installed within the filter frame.

14. The hood assembly of claim 10, wherein, in the collapsed configuration, the lip is configured to restrict rotation of the top panel in an inward direction relative to the housing, restrict movement of the top panel in a downward direction, or both, via engagement of the lip with the top panel retainer of the housing.

15. The hood assembly of claim 11, wherein the first sub-frame and the second sub-frame are configured to rotate toward one another during transition of the hood assembly from the deployed configuration to the collapsed configuration, and the first sub-frame and the second sub-frame are configured to rotate away from one another during transition of the hood assembly from the collapsed configuration to the deployed configuration.

16. A heating, ventilation, and air conditioning (HVAC) unit, comprising:

- a housing comprising a first lip;

- an intake hood assembly adjustably coupled to the housing and configured to transition between a collapsed configuration and a deployed configuration, wherein the intake hood assembly comprises:

- a top cover rotatably coupled to the housing, wherein the top cover comprises a second lip, the first lip of the housing and the second lip of the top cover are configured to engage with one another in the collapsed configuration, and the first lip of the housing and the second lip of the top cover are configured to be spaced apart from one another in the deployed configuration; and

- a filter frame rotatably coupled to the top cover and to the housing, wherein the top cover and the filter frame are configured to rotate relative to one another and relative to the housing during transition of the intake hood assembly between the collapsed configuration and the deployed configuration.

17. The HVAC unit of claim 16, wherein the filter frame comprises a first filter frame configured to support a first filter and a second filter frame configured to support a second filter.

18. The HVAC unit of claim 16, wherein the top cover is 5 configured to overlap with the filter frame and enclose the filter frame within the housing in the collapsed configuration.

19. The HVAC unit of claim 16, wherein the top cover comprises a panel portion, and the second lip of the top 10 cover extends crosswise from the panel portion.

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